Exact degree and real-point counts in a localized sector of quantum Levi–Civita connections on $(\mathbb{Z}_2)^4$

A computer-assisted classification of a marked path–triangle sector

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Abstract

We define a marked, spatially homogeneous, trace-zero path–triangle sector of bimodule quantum Levi–Civita connections for a specified differential calculus on the four-dimensional binary cube. Reversible algebraic reductions identify its real coefficient scheme with an explicit localized family of four equations in four variables. We prove that its complexification has generic geometric degree 512. At the rational metric $q^0 = (4/3, 12/11, 3/4)$, with the four scales equal to one, we prove that the entire declared fibre has length 512 and consists of 512 distinct reduced points, exactly 204 of which are real. The proof separates the generic-degree calculation, the upper bound at every finite special fibre, the construction of 512 distinct lifts, and the exact real-root count. Large integer polynomials and finite-field identities are supplied as explicit computational certificates with verification algorithms. The geometric arguments, exact identities and standard algebraic-geometric ingredients are given explicitly. The count concerns the stated marked sector, without a quotient by symmetry or a claim about the full moduli space.

Keywords: quantum Levi–Civita connections; finite geometry; scheme-level reconstruction; computer-assisted enumeration; real algebraic geometry.

1 Introduction

Quantum Levi–Civita connections on a finite differential calculus are specified by torsion, tensor metric compatibility and a star-preserving bimodule Leibniz rule. Solving these equations is a nonlinear algebraic problem even when the underlying algebra consists of functions on a small finite set. An exact enumeration must retain the calculus, the unknown tensor Leibniz map, the real structure and every open condition through elimination and reconstruction.

We study the four coordinate-flip Grassmann calculus on $(\mathbb{Z}/2)^4$, with a diagonal Lorentz-signature metric whose coefficients are spatially homogeneous and may change between two binary time slices. The family counted here is defined directly in the fixed marked connection data by a spatial path on one slice, a spatial triangle on the other, trace-zero conditions and an explicit regular open. These are coupling graphs of the connection coefficients within one sixteen-vertex calculus. The path is the minimally connected spatial support; the triangle adds one independent cycle. The two slices are coupled by the original temporal metric and star equations. The resulting completion mechanisms and a balanced tensor-block splitting are explained in Section 4.4.

The contribution is a reversible description of this original localized QLC scheme and a complete computer-assisted enumeration. Its generic complex degree is 512. At the rational metric

$q^0 = (4/3, 12/11, 3/4)$, with all scales one, its entire declared fibre is reduced with 512 complex points, precisely 204 of which are real. The inverse reconstruction produces actual star-preserving bimodule connections at all those real points. The theorem counts marked coefficient points; its support and localization are retained throughout. The explicit fibre supplies a test case for reconstruction and counting methods that must retain both the original geometric equations and the real structure while allowing the tensor Leibniz map to vary.

Three logically separate tasks are essential. The generic-degree argument supplies the upper bound at every finite special fibre. Finiteness and 512 distinct smooth characteristic-zero lifts exhaust that bound. Only then does an exact primitive-coordinate polynomial and a complete real-root certificate identify the 204 real connections. Finding 512 modular points alone would leave the completeness question unanswered. All written arguments and actual polynomial data are supplied. A single supplementary command replays the arithmetic from fresh inputs; no earlier project archive is required.

1.1 Relation to prior work

The finite-set bimodule framework, star-compatible connections and the inner description of connections are established machinery [1, 2]. In particular, [12, Definition 8.3, Proposition 8.11 and Lemma 8.17, pp. 568, 576–579 and 589–590] gives the inner normal form, torsion, full tensor metric compatibility and group component equations used here. Its group-degree argument already forces the inner correction map to vanish for the four coordinate flips. Star preservation is an additional requirement, as in [12, Lemma 8.7, p. 572]. General component equations for nonconstant Cayley data also precede this work [17, Section 6.1, Proposition 6.2 and Lemma 6.3], [14, Section 2.2 and Lemma 2.1]. The present result solves a specified restricted coefficient scheme within that framework.

Square and polygon classifications already exhibit QLC nonuniqueness; digital classifications also provide finite QLC counts [11, 14, 13]. The earlier generic square family is already given in [10, Section 5]. They depend on the specified two-form calculus as well as the first-order graph. For example, the cyclic and bit-flip squares have different two-form translation degrees and torsion equations. In the bit-flip square, $\deg(e^1 \wedge e^2) = s_1 + s_2 \neq 0$ in $(\mathbb{Z}/2)^2$; in the cyclic four-gon, $\deg(e^+ \wedge e^-) = 1 + (-1) = 0$ in $\mathbb{Z}/4$. The basic two-form therefore has different bimodule commutation with functions in these two calculi. Higher-dimensional component analysis in [16, Section 5.2, pp. 242–245] assumes cyclic orders $m_i > 4$ and distinct positive and negative directions. Setting those orders to two changes the translation-degree blocks, so it is not a formal specialization of that classification to the present input.

Uniqueness comparisons must keep their compatibility conventions and fixed data. Theorem 5.14 of [15] compares connections with the same tensor Leibniz map σ, whereas σ varies here. The inner formula below already implies uniqueness when our σ is fixed. The weak Hermitian framework of [18, Theorems 3.11 and 4.12] provides a comparison of metric and torsion predicates on star-preserving invertible bimodule pairs; it does not enumerate the marked support, traces and localized scheme considered here. We make no claim to a new connection formalism or a general uniqueness counterexample across different notions of metric compatibility.

Further existence and uniqueness results require attention to the calculus. Our one-form bimodule has zero centre: if $\omega = \sum_i a_i e^i$ is central, then $a_i(f - R_i f) = 0$ for every vertex function f, and vertex idempotents force all $a_i = 0$. The centred-calculus hypothesis of [19, Definition 3.15 and Theorem 4.24] fails here because $Z(\Omega^1) = 0$. The derivation-based calculus of [20, Definitions 2.1 and 3.12, Theorem 7.2] has $\mathfrak{g} \subseteq \mathrm{Der}(A)$, whereas $\mathrm{Der}(\mathbb{C}^{16}) = 0$: an ordinary derivation annihilates every primitive idempotent, since $D(p) = 2pD(p)$ and $1 - 2p$ is a unit. Aschieri's theorem [21, Theorem 6.7] uses the dual of the associated braided-derivation module. On our natural Hopf input $A = C(G)$, $G = (\mathbb{Z}/2)^4$, his cotriangular construction [21, Example 3.9, equations (3.25) and (3.28)] gives zero one-forms: $U = A^* = \mathbb{C}[G]$ is commutative, so its adjoint action is trivial and (3.28) reduces to the primitive-element equation, whose only solution in U is zero. Thus this construction does not give our rank-four Grassmann calculus. This last comparison concerns the natural Hopf

input and makes no assertion about every possible alternative triangular H-realization.

By contrast, [22, Theorem 5.7] does apply to our bicovariant calculus and the corresponding inverse metric $\mathfrak{g} = \sum_i k_i \xi_i \otimes \xi_i$, $k_i = h_i^{-1}$. Here h_i are the metric coefficients of Section 3, and ξ_i is the right-linear dual frame with $\langle \xi_i, e^j \rangle = \delta_{ij}$. On symmetric-output coefficients $L^i{}_{jb} = L^i{}_{bj}$, their map $\Phi_{\mathfrak{g}}$ sends L to $M_{ijb} = k_j L^i{}_{jb} + k_i L^j{}_{ib}$ and has inverse $L^i{}_{jb} = (k_b M_{ijb} + k_j M_{ibj} - k_i M_{jbi})/(2k_j k_b)$. Every denominator is a unit, so this proves invertibility here. Using the notation of Section 3, pass to a right connection by $\nabla_R = D \circ \nabla \circ *$ and write $\nabla_R e^i = \sum_{a,b} K^i_{ab} e^a \otimes e^b$. Their compatibility condition [22, Definition 5.3(ii)] becomes $k_j K^i_{jb} + k_i K^j_{ib} = \delta_{ij}(R_b k_i - k_i)$. The case $i = j = b$, edge symmetry and $K^i_{ii} = -\overline{U_i}$ force $U_i = 0$, contradicting (5). Moreover, pulling their unique connection back by the same star passage and using (2) gives the residual $E_{0ii} = -h_i(Q_i - 1)^2/4$ in (3), where $Q_i(t) = h_i(1-t)/h_i(t)$ and $i > 0$. Thus the same-metric compatibility conditions differ even without the marked trace restriction. Both Aschieri and Aschieri–Weber allow arbitrary connections in these theorems; these comparisons rely on their actual calculus and metric predicates, rather than merely on fixing a tensor braiding. Appendix A derives the rational tensor-product specialization, including the coefficient-shift cancellation, and the same-metric residual explicitly.

Our contribution is the scheme-level reconstruction, the corrected generic degree, the complete reduced rational-metric fibre and its exact real count, taken together. The inspected versions are identified in the bibliography. This statement does not assert a full four-cube moduli classification, a general elimination algorithm, or universal priority. Example 8.20 of [12, pp. 594–598] and its relevant dependencies have now been inspected in full. It treats the rank-two bit-flip square: the rectangular metric family has four nonzero complex connection parameters, and the displayed generic family has one; star preservation imposes unit modulus. The intermediate case with only one constant metric function is explicitly left without a separate classification. Section 2.3, pp. 6–7 of the inspected arXiv v1 of [11] has the same scope. These parameter families must not be identified with the complexification of our restricted real scheme. The general book machinery applies to our calculus and generates its equations, but the inspected results do not supply the coupled marked sector, its reversible reduction or the complete 512/204 enumeration. Section 4.4 also gives a concrete obstruction to obtaining our connections by restriction to coordinate squares. The detailed dependency and construction comparison is supplied in `literature/literature_comparison.pdf`. The final twenty-three-page journal version of [11] remains uninspected; no claim about its complete classification scope is inferred from its metadata or from the book's attribution of the example to that paper.

2 The statement and the proof obligations

The geometry, all equations and the exact open defining the family are given in Section 3. The four scale parameters are independent invertible base parameters; they are fixed in each counting problem. The metric ratios vary over $q_i \neq 0, 1$.

Theorem 2.1 (Degree and the exact rational-metric fibre)**.** *Let $\mathcal{Q}_{\mathrm{pt}}$ be the real localized coefficient scheme defined in Section 3, and fix all four scales to one. Let $X_{\mathbb{C}}$ be its complexification over the metric-ratio base. Then:*

(i) *Its generic fibre has* 512 *distinct geometric points and length* 512*. There is a nonempty base open on which the family is finite étale of rank* 512*.*

(ii) *At $q^0 = (4/3, 12/11, 3/4)$, its entire declared fibre is finite, reduced, and has* 512 *distinct complex points and length* 512*.*

(iii) *Exactly* 204 *points in this fibre are real. Through the explicit coefficient reconstruction, they are the star-preserving connections in the specified real marked sector at this metric.*

The word "complexification" is essential: allowing the variables of a real coefficient scheme to take complex values is not the same as imposing an antilinear star equation on arbitrary complex-valued connection coefficients. The 204 real points have the latter intended geometric interpretation

through the real scheme isomorphism proved below. Likewise positive metric parameters in the declared sign convention do not mean a positive-definite coefficient matrix.

The proof is arranged in the logically necessary order

$$\text{generic degree } 512 \implies \text{finite-fibre length cap } 512$$
$$\implies \text{exactly 512 points at } q^0 \implies 204 \text{ real points.}$$

The first arrow is supplied by a separate length lemma. The second also requires a proof that the special fibre is finite and a construction of 512 distinct characteristic-zero lifts. The last requires identifying the true coordinate polynomial and proving a complete real-root count. None of these steps is inferred merely from the number of modular solutions found by a search.

This is a computer-assisted mathematical proof. Finite polynomial certificates have the same role as long exact coefficient calculations in an appendix: the data and specified arithmetic checks are part of the proof, rather than numerical sampling. The arithmetic dependencies and exact replay scope are recorded in Section 9 and the computational supplement.

3 The original connection problem and its exact algebraic model

The sector counted below is specified here before any elimination or specialization. Put

$$Q = q_1q_2q_3, \qquad R_{\mathbb{Z}} = \mathbb{Z}[q_1, q_2, q_3, (2Q\textstyle\prod_i(q_i-1))^{-1}].$$

For the connection problem use its real base change and four independent metric scales:

$$K = (R_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{R})[\nu^{\pm1}, \lambda_1^{\pm1}, \lambda_2^{\pm1}, \lambda_3^{\pm1}].$$

The scales will subsequently all be fixed to one. They are independent base parameters, not solution variables. Write

$$A_i(0) = \lambda_i, \quad A_i(1) = q_i\lambda_i, \qquad (h_0, h_1, h_2, h_3)(t) = (\nu, -A_1(t), -A_2(t), -A_3(t)).$$

The positive metric parameter slice has $\nu, \lambda_i, q_i > 0$. No positivity assumption is used in any of the coordinate-ring identities proved in this section.

3.1 Calculus, bimodule connection, metric and star equations

Let B be an arbitrary commutative K-algebra; in particular it may have nilpotents. Set $B_{\mathbb{C}} = B[\mathrm{i}]/(\mathrm{i}^2+1)$, with conjugation fixing B, and let $\mathcal{A} = (B_{\mathbb{C}})^{(\mathbb{Z}/2)^4}$. Translation R_i flips bit i. The differential calculus and star are

$$e^i f = R_i(f)e^i, \quad df = \sum_{i=0}^{3}(R_i f - f)e^i, \quad de^i = 0, \qquad e^i \wedge e^j = -e^j \wedge e^i, \quad (e^i)^* = -e^i.$$

All functions on the cube remain in $\mathcal{A}$. Only the connection and metric coefficients are spatially homogeneous: R_1, R_2, R_3 fix them, while R_0 exchanges the two slices $t = 0, 1$. The diagonal tensor $g = \sum_i h_i e^i \otimes e^i$ is central, symmetric and real, and is nondegenerate because every h_i is a unit. Its inverse pairing is $(e^i, e^j) = \delta_{ij}R_i(h_i)^{-1}$.

A bimodule connection consists of ∇ and a bimodule map σ with

$$\nabla(f\omega) = df \otimes \omega + f\nabla\omega, \qquad \nabla(\omega f) = (\nabla\omega)f + \sigma(\omega \otimes df).$$

Here torsion freedom means $\wedge\nabla = d$; metric compatibility means $\nabla g = 0$, with the induced tensor connection; and star preservation means $\nabla(\omega^*) = \sigma D\nabla\omega$, where D is conjugate tensor flip. No Yang–Baxter equation is imposed.

Let s_i be the four coordinate translations and $\theta = \sum_i e^i$, so $df = \theta f - f\theta$. The established inner normal form [12, Proposition 8.11], obtained from the two Leibniz rules, is

$$\nabla\omega = \theta \otimes \omega - \sigma(\omega \otimes \theta) + \gamma(\omega),$$

where γ is a bimodule map. A bimodule map preserves translation degrees. The input degree of e^i has weight one, whereas every output degree $s_j + s_k$ has weight zero or two, so $\gamma = 0$. This is valid over B: vertex idempotents distinguish the translations by coefficients $0, 1, -1$, without any reducedness hypothesis.

Use the sign convention $\nabla e^i = \sum_{j,k} C^i{}_{jk} e^j \otimes e^k$. Translation degrees and torsion freedom give exactly

$$C^i{}_{ii} = U_i, \qquad C^i{}_{ij} = C^i{}_{ji} = v_{ij}, \qquad C^i{}_{jj} = w_{ij} \quad (j \neq i), \tag{1}$$

with all three-distinct-index coefficients zero. There are 56 complex coefficient values on both slices: 8 values of U, 24 of v, and 24 of w. The braiding is recovered uniquely:

$$\sigma^{il}{}_{jk} = \delta_{jl}\delta_{ki} - \mathbf{1}_{s_i+s_l=s_j+s_k} C^i{}_{jk}. \tag{2}$$

Indeed, each allowed output has exactly one possible l in the inner connection formula. Conversely this formula is a degree-preserving bimodule map and supplies both Leibniz rules.

Expanding the tensor connection gives all metric components as

$$E_{pqr} = \delta_{qr}(R_p h_r - h_r) + h_r C^r{}_{pq} + \sum_{i,j} h_i R_i(C^i{}_{jr})\sigma^{ij}{}_{pq}. \tag{3}$$

The coefficient shifts follow from moving functions past e^i. Since the two minus signs in $(e^i)^* = -e^i$ cancel,

$$D(fe^j \otimes e^k) = R_j R_k(\bar{f}) e^k \otimes e^j.$$

Thus the basis star components are

$$T^i{}_{pq} = C^i{}_{pq} + \sum_{j,k} R_j R_k(\overline{C^i{}_{jk}})\sigma^{kj}{}_{pq}. \tag{4}$$

There are 128 metric and 128 star components on both slices.

Basis star equations are sufficient on all forms, as follows. Their residual is

$$\nabla e^i + \sigma D \nabla e^i = \theta \otimes e^i - \sigma D \sigma(e^i \otimes \theta).$$

For fixed i, the inputs $e^i \otimes e^l$ have distinct translation degrees $s_i + s_l$, preserved by D and σ. Consequently basis vanishing gives $\sigma D\sigma(e^i \otimes e^l) = e^l \otimes e^i$ separately for every l. Semilinearity then yields $(\sigma D)^2 = \mathrm{id}$ on all tensors, in particular $\sigma^{-1} = D\sigma D$. For

$$\mathcal{R}(\omega) = \nabla\omega - \sigma D \nabla(\omega^*)$$

the two Leibniz rules, $(\sigma D)^2 = \mathrm{id}$, and $D(\omega^* \otimes df^*) = df \otimes \omega$ show $\mathcal{R}(f\omega) = f\mathcal{R}(\omega)$. Its vanishing on the basis therefore proves its vanishing on every form. Applying this to ω^* proves the stated star equation.

3.2 The declared real scheme and its twenty-three units

For each of the 56 coefficient values introduce independent real coordinates c_R, c_I, writing $c = c_R + \mathrm{i}c_I$. Let I be generated by the real and imaginary parts of all components (3)–(4), and of the following support and trace equations:

$$v_{13}(0) = v_{31}(0) = 0, \qquad \sum_{i=0}^{3}(U_i(t) - 1) = 0 \quad (t = 0, 1). \tag{5}$$

There are 112 real coordinate variables before taking this quotient. Define, before imposing the equations,

$$x_i(t) = v_{0i}(t), \quad y_i(t) = v_{i0}(t), \quad d_i(t) = x_i(t) + y_i(t) - 1,$$
$$X_{t,i} = x_i(1-t), \quad Y_{t,i} = y_i(t), \quad S_t = X_t^{\mathsf{T}} Y_t.$$

For two vectors write

$$E(X,Y) = X_1^2Y_2^2 - X_1^2Y_3^2 - X_2^2Y_1^2 + X_2^2Y_3^2 + X_3^2Y_1^2 - X_3^2Y_2^2,$$

and put

$$A = X_{0,1}Y_{0,2} + X_{0,2}Y_{0,1}, \quad B_0 = X_{0,2}Y_{0,3} + X_{0,3}Y_{0,2}, \quad E_1 = E(X_1, Y_1).$$

The notation B_0 here is an edge pairing, not the test algebra B. Let $\mathcal{V}$ contain both orientations of edges $12, 23$ on slice zero and $12, 13, 23$ on slice one; it has ten elements. Define

$$\Delta_{\mathrm{reg}} = \left(\prod_{i=1}^{3} d_i(0)(q_i - d_i(0))(q_i - d_i(0)^2) \right) S_0 A B_0 E_1 \prod_{v \in \mathcal{V}} v, \qquad \mathcal{Q} = \operatorname{Spec} K[c_R, c_I, N(\Delta_{\mathrm{reg}})^{-1}]/I, \tag{6}$$

where $N(z) = z\bar{z}$. We also write $\mathcal{Q}_{\mathrm{pt}} = \mathcal{Q}$ to indicate this fixed marked path–triangle sector. This is the precise regular open under study. Its factors consist of nine temporal, four completion, and ten active-edge factors, hence twenty-three in total.

For any test algebra, $N(z)$ is a unit if and only if z is a unit in $B_{\mathbb{C}}$: if $N(z)$ is invertible then $z^{-1} = \bar{z}/N(z)$; conversely conjugation preserves units. The inverse of the real element $N(z)$ in $B_{\mathbb{C}}$ is fixed by conjugation, so it lies in B; thus this equivalence also holds for the norm as an element of B. A product is invertible in a commutative ring if and only if each factor is invertible. Thus (6) inverts exactly the stated factors even before reality is proved. No pointwise active/inactive dichotomy has been used.

3.3 Eliminating the square coefficients and proving reality

Antisymmetrizing (3) in p, q cancels the quadratic terms, because the lower connection indices are symmetric:

$$\begin{aligned} E_{pqr} - E_{qpr} = {} & \delta_{qr}(R_p h_r - h_r) - \delta_{pr}(R_q h_r - h_r) \\ & + h_q R_q(C^q{}_{pr}) - h_p R_p(C^p{}_{qr}). \end{aligned} \tag{7}$$

For the six unordered pairs, $r = p, q$, and the two slices these are twenty-four linear equations with distinct w-pivots $\pm\nu$ or $\pm A_i(t)$. Their unique solution is

$$\begin{aligned} w_{i0}(t) &= -\nu x_i(1-t)/A_i(t), \\ w_{0i}(t) &= \{A_i(1-t)(1-y_i(1-t)) - A_i(t)\}/\nu, \\ w_{ij}(t) &= A_j(t) v_{ji}(t)/A_i(t) \quad (i, j \text{ spatial}, \ i \neq j). \end{aligned} \tag{8}$$

All pivots are units, so this is reversible elimination of variables in coordinate rings, without a radical operation.

For the next formulas suppress t, use a tilde for the opposite slice, and put

$$q_i(t) = A_i(1-t)/A_i(t), \quad m_i = U_i - 1, \quad M_{ii} = m_i,\ M_{ij} = v_{ij}\ (i \neq j \text{ spatial}), \quad z_i = 1 + q_i(t)(\widetilde{y}_i - 1),$$

$$\widehat{L} = \begin{pmatrix} m_0 & X^{\mathsf{T}} \\ z & M \end{pmatrix}, \qquad L^{\sharp} = \begin{pmatrix} \widetilde{m}_0 & X^{\mathsf{T}} \\ y & M \end{pmatrix}.$$

Substitution of (8) into the original components gives the following exhaustive table. Counts include both slices and duplicated components.

Indices	Count	Exact component after substitution
$p = q$	32	$E_{00r} = -\nu(\widehat{L}L^\sharp - \mathrm{I})_{0r}$, $E_{iir} = A_i(t)(\widehat{L}L^\sharp - \mathrm{I})_{ir}$.
$p \neq q$, time–space, $r = p$ or q	24	$-\nu(d_i\widetilde{x}_i - x_i)$ for $r = 0$; $A_i(t)(d_i^2 - d_i x_i + q_i(t)(x_i - 1))$ for $r = i$.
$p \neq q$, spatial, $r = p$ or q	24	$A_r(t)v_{rj}(v_{pq} + v_{qp} - 2)$, where j is the other index.
Three distinct indices	48	Zero.

For example the time–space entries follow by inserting the two displayed w-formulas into (3); the remaining off-diagonal terms have the single possible spatial mixed coefficient. For diagonal outputs the four sums are precisely the matrix product shown. These identities cover every component and prove equality of the original metric ideal and the displayed residual ideal after the reversible linear elimination, rather than just equality of field-valued zero sets.

For the star table define $W_{ij} = C^i{}_{jj}$, $\mathscr{Q} = \mathrm{I} - W$, and $d_{ij} = v_{ij} + v_{ji} - 1$. Direct use of (2) gives

Output indices	Count	Exact star component
$p = q$	32	$(\mathrm{I} - \overline{\mathscr{Q}}\mathscr{Q})_{ip}$.
$p \neq q$, $i \in \{p, q\}$	48	$v_{ij} - d_{ij}R_iR_j(\bar{v}_{ij})$, with j the other index.
Three distinct indices	48	Zero.

Summing mixed orientations yields $1 - d_{ij}R_iR_j(\bar{d}_{ij})$. Thus all d_{ij} are units. Also $\overline{\mathscr{Q}}\mathscr{Q} = \mathrm{I}$ makes $\mathscr{Q}$ invertible by the adjugate identity. The degree-zero braiding block is $\mathscr{Q}^\mathsf{T}$, and the mixed block for i, j has matrix $\left(\begin{smallmatrix} -v_{ij} & 1-v_{ji} \\ 1-v_{ij} & -v_{ji} \end{smallmatrix}\right)$ and determinant d_{ij}. This also checks invertibility of the entire braiding directly.

Now consider one time–space pair on slice zero and abbreviate

$$x = x_i(0), \quad x' = x_i(1), \quad y = y_i(0), \quad y' = y_i(1), \quad d = x + y - 1, \quad d' = x' + y' - 1, \quad q = q_i.$$

The metric residuals include

$$F_d = dx' - x, \qquad F'_d = d'x - x', \qquad H_d = d^2 - dx + q(x - 1), \qquad (q - d)x = q - d^2.$$

Our open makes $d, q - d, q - d^2$ units. Hence x and $x' = x/d$ are units, and the polynomial identity

$$dF'_d + F_d = x(dd' - 1)$$

gives $dd' = 1$. The star residuals are $G = x - d\bar{x}'$, $G' = x' - d'\bar{x}$. The identities

$$G + F_d = d(x' - \bar{x}'), \qquad G' + F'_d = d'(x - \bar{x})$$

prove x, x' real in the universal localized coordinate ring. Consequently $d = x/x'$, $d' = d^{-1}$, $y = d + 1 - x$, and $y' = d' + 1 - x'$ are real as well. This is a statement in the ring: $B_\mathbb{C} = B \oplus \mathrm{i}B$, so $c - \bar{c} = 2\mathrm{i}c_I = 0$ forces $c_I = 0$, even for nilpotent imaginary coordinates.

The temporal entries have therefore been recovered uniquely as

$$x = \frac{q - d^2}{q - d}, \quad y = \frac{d(q - 1)}{q - d}, \quad x' = x/d, \quad y' = y/d. \tag{9}$$

All four are units. They satisfy the opposite-slice conic with (q, d) replaced by (q^{-1}, d^{-1}). The replacement $z = y$, which cannot be made before this step, follows from the polynomial certificate

$$d\{1 + q(y' - 1) - y\} = -H_d + q(dd' - 1 - F_d).$$

Thus the metric core becomes $LL^\sharp = \mathrm{I}$, with

$$L = \begin{pmatrix} m_0 & X^\mathsf{T} \\ y & M \end{pmatrix}, \qquad L^\sharp = \begin{pmatrix} \widetilde{m}_0 & X^\mathsf{T} \\ y & M \end{pmatrix}.$$

A right inverse of a square matrix over a commutative ring is also a left inverse: its determinant is a unit, and the adjugate gives the inverse. The upper-right block of $LL^\sharp - L^\sharp L$ is $(m_0 - \widetilde{m}_0)X^\mathsf{T}$. Every $X_i = x_i'$ is a unit, so $m_0 = \widetilde{m}_0 =: m$. On each slice put $P_t = L = L^\sharp$; then $P_t^2 = \mathrm{I}$.

On each active spatial edge one orientation is a declared unit. The metric product in the table consequently gives $v_{ij} + v_{ji} = 2$. On the inactive path edge both orientations are zero by (5). With $H_t = \mathrm{diag}(\nu, -A_1(t), -A_2(t), -A_3(t))$, the reconstruction (8) gives the exact identity

$$\mathscr{Q} = -H_t^{-1} P_t^\mathsf{T} H_t.$$

The metric scales are real. Therefore the degree-zero star equations give $P_t \bar{P}_t = \mathrm{I}$. Combined with $P_t^2 = \mathrm{I}$, they imply $\bar{P}_t = P_t$, with the explicit polynomial certificate

$$\bar{P} - P = P(P\bar{P} - \mathrm{I}) - (P^2 - \mathrm{I})\bar{P}.$$

All U, v, and then all w, are real in the coordinate ring. The trace equations are exactly $\mathrm{tr}\, P_t = 0$.

3.4 The temporal chart and the block problem

Write $u_i = y_i(0)$, and set

$$\alpha_i = q_i - (u_i - 1)^2, \qquad \beta_i = q_i + u_i - 1.$$

The inverse changes of coordinates, together with every required unit identity, are

$$\begin{aligned} u_i &= \frac{d_i(q_i-1)}{q_i - d_i}, & \beta_i &= \frac{q_i(q_i-1)}{q_i - d_i}, & \alpha_i &= \frac{q_i(q_i-1)(q_i - d_i^2)}{(q_i-d_i)^2}, \\ d_i &= \frac{q_i u_i}{\beta_i}, & q_i - d_i &= \frac{q_i(q_i-1)}{\beta_i}, & q_i - d_i^2 &= \frac{q_i(q_i-1)\alpha_i}{\beta_i^2}. \end{aligned} \tag{10}$$

They prove isomorphism of the two temporal localizations, including nonreduced test algebras. The two slice vectors become exactly

$$X_{0,i} = \frac{\alpha_i}{q_i u_i}, \quad Y_{0,i} = u_i, \qquad X_{1,i} = \frac{\alpha_i}{\beta_i}, \quad Y_{1,i} = \frac{\beta_i}{q_i}. \tag{11}$$

Every entry is a unit, and

$$z_i = X_{t,i} Y_{t,i} = \alpha_i / q_i, \qquad S = S_0 = S_1 = \sum_i z_i.$$

Thus the intermediate block problem has

$$P_t = \begin{pmatrix} m & X_t^\mathsf{T} \\ Y_t & M_t \end{pmatrix}, \qquad M_0 = \begin{pmatrix} a_0 & p_0 & 0 \\ 2-p_0 & b_0 & s_0 \\ 0 & 2-s_0 & c_0 \end{pmatrix}, \quad M_1 = \begin{pmatrix} a_1 & p_1 & r_1 \\ 2-p_1 & b_1 & s_1 \\ 2-r_1 & 2-s_1 & c_1 \end{pmatrix},$$

subject to $P_t^2 = \mathrm{I}$, $\mathrm{tr}\, P_t = 0$, and precisely the twenty-three units already declared. The subscripts on a_t, b_t, c_t are slice labels, unrelated to the metric scales.

Lemma 3.1. *Over any commutative ring with $2, S = X^\mathsf{T}Y$ invertible, suppose $MY = -mY$, $X^\mathsf{T}M = -mX^\mathsf{T}$, and $\mathrm{tr}\, M = -m$. Then the block matrix $P = \left(\begin{smallmatrix} m & X^\mathsf{T} \\ Y & M \end{smallmatrix}\right)$ is an involution if and only if*

$$F = m^2 + S - 1 = 0, \qquad G = \mathrm{tr}(M^2) - m^2 - 2 = 0.$$

This is equality of equation ideals after the linear and trace equations.

Proof. Put $\Pi = YX^{\mathsf{T}}/S$. The free module of rank three splits as $AY \oplus \ker X^{\mathsf{T}}$. The second summand is a projective module of rank two; the linear equations make M act by $-m$ on the first summand and by a trace-zero endomorphism N on the second. Locally a trace-zero matrix $\left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ squares to $(a^2+bc)\mathrm{I}$, half the trace of its square. These endomorphism identities glue. It follows that

$$M^2 + YX^{\mathsf{T}} - \mathrm{I} = \frac{G}{2}(\mathrm{I} - \Pi) + F\Pi. \tag{12}$$

The other blocks of $P^2 - \mathrm{I}$ are F and the assumed linear residuals. Conversely, if $R_M = M^2 + YX^{\mathsf{T}} - \mathrm{I}$, then $G = \operatorname{tr} R_M - F$. Both inclusions of ideals follow in the universal quotient by the linear and trace equations. □

The localization at S is substantive. For example $m = 1$, $X = (1,1,1)^{\mathsf{T}}$, $Y = (1,1,-2)^{\mathsf{T}}$, and $M = \left(\begin{smallmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{smallmatrix}\right)$ have $S = 0$, the linear and trace equations, and $F = G = 0$, but $M^2 + YX^{\mathsf{T}} - \mathrm{I} = YX^{\mathsf{T}} \neq 0$. This tests the lemma, without asserting that the example is in our selected support sector.

3.5 Path completion and its three scalar equations

For this subsection X, Y are the path vectors. Abbreviate

$$A = X_1Y_2 + X_2Y_1, \quad B = X_2Y_3 + X_3Y_2, \quad z_i = X_iY_i,$$

$$C = A(z_2 + z_3) + B(z_1 + z_2), \qquad L = ABm + C, \qquad H_p = AB + 4z_2.$$

The six linear equations $M_0Y = -mY$, $X^{\mathsf{T}}M_0 = -mX^{\mathsf{T}}$ have the unique solution

$$\begin{aligned} p_0 &= \frac{2X_2Y_1}{A}, & s_0 &= \frac{2X_3Y_2}{B}, & & \\ a_0 &= -m - \frac{2z_2}{A}, & c_0 &= -m - \frac{2z_2}{B}, & b_0 &= -m - \frac{2z_1}{A} - \frac{2z_3}{B}. \end{aligned} \tag{13}$$

For uniqueness, take X_1 times the first right residual minus Y_1 times the first left residual. It is $Ap_0 - 2X_2Y_1$. The analogous residual at vertex three gives $Bs_0 - 2X_3Y_2$. Since A, B, Y_i are units, these determine the two edges and all diagonal entries. Direct substitution verifies the remaining linear residuals.

After this completion put $F_1 = \operatorname{tr} M_0 + m$ and $F_2 = \operatorname{tr}(M_0^2) - m^2 - 2$. Expansion gives

$$\begin{aligned} F_1 &= -2L/(AB), \\ AB(F_2 - 2F) + 2CF_1 + 2SH_p &= 0. \end{aligned} \tag{14}$$

For example $p_0(2 - p_0) = 4z_1z_2/A^2$ and $s_0(2 - s_0) = 4z_2z_3/B^2$; these and the three diagonal squares give the second identity. Explicit inverse generator transformations are

$$F_2 = 2F + \frac{4CL}{(AB)^2} - \frac{2SH_p}{AB}, \qquad H_p = -\frac{AB(F_2 - 2F) + 2CF_1}{2S}.$$

Every denominator is a declared unit. Hence $(F, F_1, F_2) = (F, L, H_p)$ as ideals, and the trace-zero path involution is equivalent to $F = L = H_p = 0$.

3.6 Triangle completion and an explicit invariant residual

Now X, Y are the triangle vectors. The unknowns $(a_1, b_1, c_1, p_1, r_1, s_1)$ are determined by the three right linear equations, the first two left equations, and the trace. Their matrix and right-hand side

are

$$\mathcal{T} = \begin{pmatrix} Y_1 & 0 & 0 & Y_2 & Y_3 & 0 \\ 0 & Y_2 & 0 & -Y_1 & 0 & Y_3 \\ 0 & 0 & Y_3 & 0 & -Y_1 & -Y_2 \\ X_1 & 0 & 0 & -X_2 & -X_3 & 0 \\ 0 & X_2 & 0 & X_1 & 0 & -X_3 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} -mY_1 \\ -2Y_1 - mY_2 \\ -2Y_1 - 2Y_2 - mY_3 \\ -mX_1 - 2X_2 - 2X_3 \\ -mX_2 - 2X_3 \\ -m \end{pmatrix}.$$

Expansion of its determinant gives $\det \mathcal{T} = Y_3 E(X, Y)$, which is a unit on our open. Thus the solution is unique over every test ring. The omitted third left equation follows from

$$X^{\mathsf{T}}(MY + mY) = (X^{\mathsf{T}}M + mX^{\mathsf{T}})Y$$

and the unit Y_3. Define polynomial numerators, in the displayed coordinate order, by

$$(\Theta_a, \Theta_b, \Theta_c, \Theta_p, \Theta_r, \Theta_s)^{\mathsf{T}} = \mathrm{adj}(\mathcal{T})\mathbf{b}/Y_3.$$

Each component of the numerator is divisible by Y_3 in the integer polynomial ring; this can equivalently be checked by the determinant expansion just specified. Thus this is an explicit polynomial definition, not a choice of a rational solution. The completed matrix is $M_1 = \mathcal{N}/E$, where

$$\mathcal{N} = \begin{pmatrix} \Theta_a & \Theta_p & \Theta_r \\ 2E - \Theta_p & \Theta_b & \Theta_s \\ 2E - \Theta_r & 2E - \Theta_s & \Theta_c \end{pmatrix}.$$

Let $G_{\mathrm{num}} = \mathrm{tr}(\mathcal{N}^2) - (m^2 + 2)E^2$ and $Q_\triangle = [m^2]G_{\mathrm{num}}$. One has the polynomial identity

$$G_{\mathrm{num}} = Q_\triangle F - 2SH_{\mathrm{tri}}(X, Y, m). \tag{15}$$

Here $Q_\triangle$ is independent of m, and the following orbit formulas give an explicit residual $H_{\mathrm{tri}} = H_0 + mH_1$. Write $z_i = X_iY_i$, $a_{ij} = X_iY_j + X_jY_i$. For a monomial f, $\mathcal{O}(f)$ denotes the sum of its distinct simultaneous S_3-images, permuting vertex and unordered-edge labels. Set

$$\begin{aligned} H_1 &= 16\mathcal{O}(z_1z_2a_{12}) + 8\mathcal{O}(z_1a_{12}a_{23}) - 4\mathcal{O}(a_{12}^3) + 4\mathcal{O}(a_{12}^2a_{13}), \\ H_0 &= 16\mathcal{O}(z_1^2z_2) + 16\mathcal{O}(z_1^2a_{23}) + 32z_1z_2z_3 \\ &\quad - 4\mathcal{O}(z_1z_2a_{12}^2) + 16\mathcal{O}(z_1z_2a_{13}) - 4\mathcal{O}(z_1a_{12}a_{13}a_{23}) \\ &\quad + 8\mathcal{O}(z_1a_{12}a_{23}) + \mathcal{O}(a_{12}^4) + 4\mathcal{O}(a_{12}^3) \\ &\quad - 2\mathcal{O}(a_{12}^2a_{13}^2) - 4\mathcal{O}(a_{12}^2a_{13}) + 16a_{12}a_{13}a_{23}. \end{aligned} \tag{16}$$

The identity (15) is an identity after the bilinear substitution, with no algebraically independent extra bilinear variables. In particular

$$4z_1z_2z_3 + a_{12}a_{13}a_{23} - z_1a_{23}^2 - z_2a_{13}^2 - z_3a_{12}^2 = 0$$

holds identically. Equations (15) and (16) are directly verified by replacing each Θ with the adjugate definition and expanding; they are finite integer polynomial identities. The accompanying executable certificate performs exactly this expansion, including all six numerators, all seven linear/trace constraints, and the displayed orbit polynomial.

Since $2, S, E$ are units,

$$(F, \mathrm{tr}(M_1^2) - m^2 - 2) = (F, H_{\mathrm{tri}})$$

after completion. Indeed $E^2(\mathrm{tr}(M_1^2) - m^2 - 2) = Q_\triangle F - 2SH_{\mathrm{tri}}$ exhibits both generator transformations. Lemma 3.1 now gives equivalence with the full trace-zero triangle involution.

3.7 The four-variable family, its cleared equations and exact open

Use (11), and use A, B for the path pairings from now on. The resulting family has four coordinates (u_1, u_2, u_3, m) and exactly four rational equations:

$$\boxed{\begin{aligned} H_p &= AB + 4z_2 = 0, \\ L &= ABm + A(z_2 + z_3) + B(z_1 + z_2) = 0, \\ F &= m^2 + \sum_i z_i - 1 = 0, \\ H_{\mathrm{tri}} &= H_0(z, a_1) + mH_1(z, a_1) = 0. \end{aligned}} \tag{17}$$

In the last equation $(a_1)_{ij} = X_{1,i}Y_{1,j} + X_{1,j}Y_{1,i}$. Denote this localized scheme over $R_{\mathbb{Z}}$ by $\mathcal{X}$. The open is intrinsically defined by the temporal, completion and active edge units above. The next formulas give its exact polynomial inventory.

Put

$$\overline{A} = q_2\alpha_1 u_2^2 + q_1\alpha_2 u_1^2, \qquad \overline{B} = q_3\alpha_2 u_3^2 + q_2\alpha_3 u_2^2, \qquad \overline{S} = Q\sum_i \alpha_i/q_i.$$

Then $A = \overline{A}/(q_1q_2u_1u_2)$ and $B = \overline{B}/(q_2q_3u_2u_3)$. The full inventory is

Intrinsic units	Cleared polynomial factors	Count
Temporal chart	u_i, α_i, β_i $(i = 1, 2, 3)$	9
S	$\overline{S}$	1
A, B	$\overline{A}, \overline{B}$	2
Four path entries	$q_1\alpha_2u_1^2,\ q_2\alpha_1u_2^2, q_2\alpha_3u_2^2,\ q_3\alpha_2u_3^2$	4
E_1, six triangle entries	The seven polynomials defined immediately below	7

For example

$$p_0 = 2q_1\alpha_2u_1^2/\overline{A}, \qquad 2 - p_0 = 2q_2\alpha_1u_2^2/\overline{A},$$

and the other two path entries have the analogous formulas with $\overline{B}$. Some listed factors are redundant on this open. They are retained, so no change of sector is hidden in simplification.

For a triangle polynomial $J = \sum c_{x,y,k}X^xY^ym^k$, define

$$b_i(J) = \max_{c_{x,y,k}\neq 0}(x_i - y_i), \quad c_i(J) = \max_{c_{x,y,k}\neq 0} y_i, \qquad \mathcal{C}_1(J) = \left(\prod_i \beta_i^{b_i(J)}q_i^{c_i(J)}\right)J(X_1, Y_1, m).$$

Termwise use of (11) shows that this is a polynomial: the exponents are $b_i + y_i - x_i \geq 0$ and $c_i - y_i \geq 0$. All clearing monomials are units on the temporal chart. The remaining seven factors are precisely

$$\mathcal{C}_1(E), \quad \tfrac{1}{2}\mathcal{C}_1(\Theta_p), \quad \tfrac{1}{2}\mathcal{C}_1(2E - \Theta_p), \quad -\tfrac{1}{2}\mathcal{C}_1(\Theta_r), \quad \tfrac{1}{2}\mathcal{C}_1(2E - \Theta_r), \quad \tfrac{1}{2}\mathcal{C}_1(\Theta_s), \quad -\tfrac{1}{2}\mathcal{C}_1(2E - \Theta_s).$$

The constant signs and factors 2 are units. Since the six completed triangle entries are Θ/E or $(2E - \Theta)/E$, the polynomial and intrinsic opens agree over every test algebra.

For a polynomial rather than rational presentation, adjoin five graph coordinates $\xi, \eta, c_{12}, c_{13}, c_{23}$. Their defining equations are

$$\begin{aligned} q_1q_2u_1u_2\xi - \overline{A} = 0, \qquad & q_2q_3u_2u_3\eta - \overline{B} = 0, \\ q_iq_j\beta_i\beta_j(c_{ij} - X_{1,i}Y_{1,j} - X_{1,j}Y_{1,i}) = 0 \qquad & (ij = 12, 13, 23). \end{aligned} \tag{18}$$

Their coefficients on the new variables are units, so these variables adjoin unique regular sections. The other four polynomial generators are

$$\begin{aligned} & q_2(\xi\eta + 4\alpha_2/q_2), \\ & Q\{\xi\eta m + \xi(z_2 + z_3) + \eta(z_1 + z_2)\}, \\ & Q(m^2 + z_1 + z_2 + z_3 - 1), \\ & -Q^4\{H_0(z, c) + mH_1(z, c)\}. \end{aligned} \tag{19}$$

They are integer polynomials; the base units clear all denominators. Together (18) and (19) give the nine-generator presentation of the same localized four-variable family. The complete coefficients of these generators and of all twenty-three factors are supplied, and the symbolic certificate checks the displayed definitions against every coefficient.

3.8 Binding the ten-equation source and the centre equations

An alternative polynomial presentation keeps the six triangle entries $(a, b, c, p, r, t) = (a_1, b_1, c_1, p_1, r_1, s_1)$. Put

$$M = \begin{pmatrix} a & p & r \\ 2-p & b & t \\ 2-r & 2-t & c \end{pmatrix}, \qquad \mathcal{K} = \overline{A}(q_3\alpha_2 + q_2\alpha_3)u_3 + \overline{B}(q_2\alpha_1 + q_1\alpha_2)u_1.$$

The ten literal source equations used in the exact computation are the following polynomials, in the coordinate order $(u_1, u_2, u_3, m, a, b, c, p, r, t)$:

$$\begin{aligned}
&Q(m^2-1) + \sum_i q_{i+1}q_{i+2}\alpha_i, \\
&\overline{AB} + 4Q\alpha_2 u_1 u_2^2 u_3, \qquad m\overline{AB} + u_2\mathcal{K}, \\
&\sum_{j=1}^{3}(M_{ij} + m\delta_{ij})\beta_j q_{j+1}q_{j+2} \quad (i = 1, 2, 3), \\
&\sum_{j=1}^{3}(M_{ji} + m\delta_{ij})\alpha_j\beta_{j+1}\beta_{j+2} \quad (i = 1, 2), \\
&a + b + c + m, \\
&a^2 + b^2 + c^2 + 2\{p(2-p) + r(2-r) + t(2-t)\} - m^2 - 2.
\end{aligned} \tag{20}$$

The indices on q, α, β are cyclic modulo three. To verify scheme equivalence, set $D_p = q_1q_2^2q_3u_1u_2^2u_3$, a temporal unit. The first three source rows are exactly QF, D_pH_p, D_pL. The next three are $Q(MY_1 + mY_1)_i$; the next two are $(\beta_1\beta_2\beta_3)(X_1^{\mathsf{T}}M + mX_1^{\mathsf{T}})_i$. The last two are the trace and trace-square residuals. All multipliers are units. The triangle linear system is invertible on our declared $Y_{1,3}E_1$ open. The omitted left equation and the inverse transformations in (15) therefore prove equivalence with (17) as localized schemes. A separate exact integer-polynomial comparison checks every literal coefficient of (20); it is not a modular comparison.

One must also specify the representative used to eliminate m. Since $z_2 = \alpha_2/q_2$ is a unit, define

$$m_* = C/(4z_2).$$

The exact identity

$$L = AB(m - m_*) + m_*H_p \tag{21}$$

gives $(H_p, L) = (H_p, m - m_*)$ on the path-completion open. The expression $-C/(AB)$, obtained from $L = 0$ alone, agrees with m_* modulo H_p; they are not identical rational functions. Indeed their difference is $CH_p/(4z_2AB)$.

At $q^0 = (4/3, 12/11, 3/4)$, eliminating $m = m_*$ gives the three primitive integer centre numerators P_p, P_F, P_T, of respective u_1-degrees $2, 6, 10$. Their canonical scalar identities are (42)–(44). Every multiplier is a declared unit. The binding checker reconstructs these rational residuals over $\mathbb{Q}[u_1, u_2, u_3]$, checks zero remainder after division by the complete supplied polynomial, and verifies all multiplier factors. Thus these are equalities of localized ideals over $\mathbb{Q}$.

3.9 Inverse reconstruction and the scheme-theoretic conclusion

Theorem 3.2. *The real scheme $\mathcal{Q}$ of* (6) *is isomorphic over* $\operatorname{Spec} K$ *to* $\mathcal{X} \times_{\operatorname{Spec} R_{\mathbb{Z}}} \operatorname{Spec} K$. *All imaginary connection coordinates vanish in its coordinate ring. The isomorphism is natural over arbitrary commutative K-algebras, including nonreduced algebras. It identifies actual real star-preserving connections with the real points of $\mathcal{X}$; its complexification is the complexification of this real algebraic model.*

Proof. The forward construction has already been carried out in the universal localized coordinate ring. It reads $d_i = x_i(0) + y_i(0) - 1$, $u_i = y_i(0)$, $m = U_0(0) - 1$, and the spatial M_t from the original coefficients. The component tables and unit pivots prove reality, common m, the two block involutions and their traces. The temporal coordinate inverse, unique path and triangle completions, and the two residual identities prove all four equations (17). The explicit factor inventory proves that this map lands in the exact declared open.

Conversely, start with an $\mathcal{X}$-point over a K-algebra. Set $d_i = q_i u_i/\beta_i$, and reconstruct all time–space entries by (9); equivalently use the two vectors (11). Construct M_0 by (13) and $M_1 = \mathcal{N}/E$ by the adjugate formula. The inverse ideal identities in (14) and (15), with Lemma 3.1, give both trace-zero block involutions. Set

$$U_0(0) = U_0(1) = 1 + m, \qquad U_i(t) = 1 + (M_t)_{ii}, \qquad v_{ij}(t) = (M_t)_{ij} \quad (i \neq j \text{ spatial}),$$

recover all w using (8), and set all imaginary coordinates to zero. This reconstructs all 56 complex coefficient values without any root choice.

The lower-index symmetry verifies torsion freedom. Formula (2) and the inner connection formula give the bimodule connection. The temporal identities verify all time–space metric residuals; the active edge sums equal two and the inactive pair is zero; the block involutions verify every metric-core residual. The exhaustive metric table therefore proves $E_{pqr} = 0$ for every component.

For star preservation, on an active spatial edge $d_{ij} = 1$ and the entries are real; on the inactive edge both entries vanish. On a temporal edge (9) gives $x = dx'$, $y = dy'$ and their opposite-slice counterparts. Finally $\mathscr{Q} = -H_t^{-1} P_t^{\mathsf{T}} H_t$ is real with $\mathscr{Q}^2 = \mathrm{I}$, so the zero-degree star block vanishes. The exhaustive star table verifies every ${T^i}_{pq} = 0$; the argument following (4) extends this to all forms and proves invertibility of σ. All original support, trace, and regular-open conditions are satisfied.

The forward and backward formulas are inverse: the temporal coordinate maps are inverse by (10); the path and triangle entries are unique solutions with unit pivots; and the eliminated square coefficients have unique unit-leading linear equations. They commute with arbitrary maps of test algebras. They thus define inverse maps of localized coordinate rings, proving the asserted scheme isomorphism, not merely a bijection of classical points. □

In particular, after all four scales are fixed to one and $q^0 = (4/3, 12/11, 3/4)$, a real point counted later is an actual star-preserving connection in this precise sector. The complex count is the geometric count in the complexification of its real coordinate scheme. It does not assert that arbitrary complex points, with an independently imposed analytic conjugation operation, are star-preserving connections. The omitted exceptional parameters, other supports, other traces, completion-degenerate charts, and unmarked quotients are not part of the defined counting problem.

4 Preservation of the original QLC problem and the meaning of the sector

Theorem 3.2 gives the scheme isomorphism. The following tables record how its two maps transport the geometric requirements: fixed data and sector restrictions, all inverted factors, and the original QLC residuals. The final subsection explains the geometric meaning of the chosen support and trace. The bimodule conventions are those of [1, 2].

4.1 Fixed data, imposed restrictions and derived consequences

The base space has all sixteen vertices of $(\mathbb{Z}/2)^4$ and the four specified translation directions throughout. Spatial homogeneity restricts coefficients, not the algebra of functions or the differential calculus. The ordered frame and the two labelled slices are retained. The following distinctions prevent a consequence of the equations from being silently substituted for an additional sector assumption.

Item	Status before reduction	What is established afterwards
Calculus and metric	Fixed as in Section 3; the four scales and q_i are base parameters.	All bimodule, metric and star calculations use the same shifts and sign convention.
Spatial homogeneity	Imposed on coefficients on each slice.	The resulting components still define connections on the full sixteen-point algebra.
Spatial support	Both 13 orientations vanish on slice zero; the ten specified orientations are units.	Exactly the path and triangle coupling patterns remain; active pairs have sum two.
Trace	$\sum_i (U_i(t) - 1) = 0$ is imposed on each slice.	It becomes $\operatorname{tr} P_t = 0$; it is not deduced from torsion or metric compatibility.
Reality and star	The original star equations are imposed in real and imaginary coordinates.	Every imaginary coefficient vanishes in the localized coordinate ring; this is proved, not assumed.
Regular open	The twenty-three intrinsic factors in (6) are inverted.	All subsequent denominators are units on precisely that open. Its excluded strata are not proved empty.
Bimodule invertibility	Both Leibniz rules and star preservation are required.	$\sigma^{-1} = D\sigma D$ follows; no Yang–Baxter equation is added.

The independent assumptions in this table define the stated restricted problem. In particular the trace, support and regularity choices are not consequences of being a QLC. Conversely none is dropped when the four-variable scheme is counted. Fixing the four scales to one selects a metric in that problem; it is not a gauge quotient.

4.2 An explicit ledger for every inverted factor

There are two kinds of units. The fixed base already inverts $2, q_i, q_i - 1, \nu, \lambda_i$ and hence every $h_i(t)$. The connection sector additionally inverts the twenty-three factors of Δ_{reg}. The inventory below refers to the latter only. It retains even the redundant path factors.

The temporal exchange is proved by all six identities in (10). Thus the nine units $(d_i, q_i - d_i, q_i - d_i^2)$ are exchanged for (u_i, β_i, α_i), with inverse formulas in both directions. In particular $q_i - d_i$ corresponds to the inverse of β_i times a base unit; it need not be represented by a new polynomial factor beyond the temporal chart.

For a triangle numerator J write

$$W_J = \prod_i \beta_i^{b_i(J)} q_i^{c_i(J)}, \qquad \mathcal{C}_1(J) = W_J J(X_1, Y_1, m).$$

Each W_J is a unit of the temporal localization. Denote the six active triangle numerators by

$$J_p = \Theta_p, \quad J_{2-p} = 2E - \Theta_p, \quad J_r = \Theta_r, \quad J_{2-r} = 2E - \Theta_r, \quad J_s = \Theta_s, \quad J_{2-s} = 2E - \Theta_s.$$

These are the exact adjugate polynomials of the triangle system, evaluated on its slice. The factors actually supplied in the coefficient files, and their pullbacks, are as follows.

No.	Polynomial factor	Exact connection-level interpretation
1–3	u_i	$u_i = d_i(q_i-1)/(q_i-d_i)$, for $i=1,2,3$.
4–6	α_i	$\alpha_i = q_i(q_i-1)(q_i-d_i^2)/(q_i-d_i)^2$.
7–9	β_i	$\beta_i = q_i(q_i-1)/(q_i-d_i)$.
10	$\overline{S}$	$\overline{S} = QS_0 = QS_1$.
11	$\overline{A}$	$\overline{A} = q_1q_2u_1u_2A$.
12	$\overline{B}$	$\overline{B} = q_2q_3u_2u_3B$.
13	$q_1\alpha_2u_1^2$	$\overline{A}\,p_0/2$.
14	$q_2\alpha_1u_2^2$	$\overline{A}\,(2-p_0)/2$.
15	$q_2\alpha_3u_2^2$	$\overline{B}\,s_0/2$.
16	$q_3\alpha_2u_3^2$	$\overline{B}\,(2-s_0)/2$.
17	$\mathcal{C}_1(E)$	W_EE_1.
18	$\mathcal{C}_1(J_p)/2$	$W_{J_p}E_1p_1/2$.
19	$\mathcal{C}_1(J_{2-p})/2$	$W_{J_{2-p}}E_1(2-p_1)/2$.
20	$-\mathcal{C}_1(J_r)/2$	$-W_{J_r}E_1r_1/2$.
21	$\mathcal{C}_1(J_{2-r})/2$	$W_{J_{2-r}}E_1(2-r_1)/2$.
22	$\mathcal{C}_1(J_s)/2$	$W_{J_s}E_1s_1/2$.
23	$-\mathcal{C}_1(J_{2-s})/2$	$-W_{J_{2-s}}E_1(2-s_1)/2$.

For instance, the factor in row 20 differs from the intrinsic active entry r_1 by the already invertible $-W_{J_r}E_1/2$. The same statement holds for every active-entry row. The temporal inverse formulas first establish the temporal chart, rows 10–12 establish S, A, B, and row 17 establishes E_1. Rows 13–16 and 18–23 then establish exactly the active-edge units. This order proves the equivalence of the opens without circular use of a denominator.

Before reality has been established, the original real scheme uses $N(\Delta_{\mathrm{reg}})$, not an unsupported notion of a complex number being nonzero. The identity $z^{-1} = \bar{z}/N(z)$ proves the equivalence of the unit conditions over arbitrary real test algebras. After reality, $N(\Delta_{\mathrm{reg}}) = \Delta_{\mathrm{reg}}^2$, and localizing at a square is identical to localizing at its factor: the inverse of the factor is $\Delta_{\mathrm{reg}}(\Delta_{\mathrm{reg}}^2)^{-1}$. Thus no reducedness assumption is concealed in the use of an open set.

For completeness, all classes of division in the reconstruction are listed here; a determinant used to prove a unit is not an extra localizer.

Operation	Divisors	Justification
Eliminate the 24 square coefficients	$\nu, A_i(t)$	Fixed metric units.
Temporal coordinate changes	$d_i, q_i-d_i, q_i, \beta_i$	Temporal unit identities above.
Split the rank-three block	$2, S$	Base and row 10.
Path completion	$A, B, Y_{0,i}$	Rows 11–12 and temporal chart.
Triangle completion	$Y_{1,3}E_1$	Temporal chart and row 17.
Exchange scalar generators	$2, S, E_1^2, AB, (AB)^2$	The displayed inverse syzygies use units already listed.
Adjoin graph coordinates	$q_iq_ju_iu_j$, $q_iq_j\beta_i\beta_j$	Base and temporal chart.
Eliminate m	$4z_2$, AB	$z_2 = \alpha_2/q_2$ and rows 11–12.
Centre numerator binding	u_i, α_2, β_i and nonzero rationals	Exactly the factors in (42)–(44).

The later primitive-coordinate circuit introduces further written denominators only after proving they are units in the already identified finite coordinate algebra. Its inverse certificates and the checks of the same twenty-three original factors are part of the centre proof; they do not define a smaller connection sector.

4.3 Both directions of the preservation argument

To specify exactly what is transported, put $R_t = P_t^2 - I$. The real reconstructed coefficients satisfy the following identities *before* the equations $R_t = 0$ are imposed:

$$E_{ppr}(t) = -h_p(t)(R_t)_{pr}, \qquad T^i{}_{pp}(t) = -\frac{h_p(t)}{h_i(t)}(R_t)_{pi}, \tag{22}$$

and all off-diagonal-output metric and star components vanish. Here the temporal entries are the exact conic formulas, both blocks share m, and active spatial pairs have sum two; the inactive pair is zero. These are statements of rational identities in the metric and temporal localization with otherwise free spatial block entries. They are stronger than evaluating the residuals at selected solutions. The independent direct tensor checker described in the verification section recomputes these identities without importing the component-table or block-completion programs.

Original requirement	Forward implication	Reverse verification
Both Leibniz rules	Translation-degree preservation forces the inner formula with no γ and determines σ.	The recovered σ is a bimodule map, and the same inner formula supplies the rules on every vertex idempotent, hence every function.
Torsion zero	The allowed lower indices become symmetric; all three-distinct coefficients vanish.	Reconstruct ${C^i}_{jk} = {C^i}_{kj}$; then $\wedge\nabla e^i = de^i = 0$ and torsion is left linear.
Metric compatibility	Unit-leading antisymmetric components eliminate each w once; the exhaustive table yields the conics and block equations.	Conic identities cancel all off-diagonal outputs, and (22) cancels all diagonal outputs.
Star preservation	Mixed residuals first prove temporal reality and reciprocity; the degree-zero matrix identity then proves all reality.	All coefficients are real; the original star residuals vanish by (22), and $\sigma D\sigma D = I$ extends basis preservation to every form.
Support and trace	Active units force the edge sums; the trace becomes $\operatorname{tr} P_t = 0$ without elimination of a branch.	Path and triangle formulas give the exact zero/unit support, both trace constraints and the same m.
Open and multiplicity	The norm localization and every unit pivot are retained; no radical is taken.	The factor ledger and inverse formulas give the original localization over every test algebra.

The full forward chain has specific algebraic witnesses. The 24 antisymmetric metric equations have distinct unit pivots; the temporal relations give $dd' = 1$ and force each imaginary coefficient to vanish; the two matrix inverses give the common m; and the path and triangle syzygies give inverse transformations of equation ideals. Uniqueness of both completions and of the eliminated w values then proves that forward and reverse maps compose to the identity on every coefficient. These are the steps proved in Theorem 3.2, not additional assumptions on the counted points.

Equivalently, for every commutative K-algebra B there is a natural bijection

$$\mathcal{X}(B) \;\longleftrightarrow\; \left\{ \begin{array}{c} (\nabla,\sigma) \text{ satisfying the original two Leibniz rules,} \\ \wedge\nabla = d, \quad \nabla g = 0, \quad \nabla(\omega^*) = \sigma D \nabla \omega, \\ \text{and precisely the homogeneity, support, trace and units above} \end{array} \right\}.$$

On the left the base change from $R_{\mathbb{Z}}$ to K is understood. This statement applies to dual numbers and other nonreduced algebras: it preserves infinitesimal multiplicities, not just field-valued solutions. Consequently each of the 204 real points later counted reconstructs an actual pair (∇,σ) satisfying the original QLC conditions. The 512 complex points belong to the complexification of this real coefficient scheme; no analytic conjugation condition is imposed independently on its arbitrary complex points.

4.4 Why this path–triangle sector is geometrically interesting

Define the *spatial coupling graph* on the three marked directions $\{1,2,3\}$ by the active v_{ij} pairs. This is a graph of connection coefficients, not a replacement for the sixteen-vertex base graph of the differential calculus. Its path P_3 at slice zero is minimally connected on those three directions; its triangle K_3 at slice one adds exactly one edge and has one independent graph cycle. That is a controlled change from tree mixing to cyclic mixing within one unchanged geometry. A cycle of this coefficient graph alone proves no curvature or holonomy statement.

The choice changes the completion mechanism in a precise way. The path has five spatial unknowns. Its left and right eigenvector equations determine them, so trace zero becomes a remaining compatibility equation $L = 0$. The triangle has six spatial unknowns. The eigenvector equations have a universal dependence, and the trace completes the invertible six-by-six system with determinant $Y_{1,3}E_1$. The units A, B, E_1, S specify where these mechanisms are reversible. Their complements remain excluded strata, whose QLC classification is a separate problem.

The slices are not chosen independently: the original temporal metric and star equations force the same m and $X_{0,i}Y_{0,i} = X_{1,i}Y_{1,i} = \alpha_i/q_i$. Thus a path solution and a triangle solution cannot simply be paired arbitrarily. Both supports participate in the same reconstructed four-dimensional bimodule connection. In fact every temporal–spatial coefficient $v_{0i}(t)$ is a unit by (9). Corollary 10.2 consequently excludes an ordinary coordinate-product construction, even after a permutation of the four direction labels.

There is also an obstruction to restricting these connections to a coordinate square. Let π be the standard calculus projection to that face: fix the other bits in functions, retain directions $(0, i)$ and kill the other one-forms. For a discarded spatial direction $j \neq i$, (8) gives the exact coefficient

$$\mathrm{coeff}_{e^0 \otimes e^0}((\pi \otimes \pi)\nabla e^j) = C^j{}_{00} = w_{j0}(t) = -\frac{\nu x_j(1-t)}{A_j(t)}.$$

These coefficients depend only on t, so fixing the other bits leaves this formula unchanged. The displayed coefficient is a unit by (9), so the kernel of the quotient on forms is not preserved by the connection. Thus there is no induced connection on that coordinate-square quotient. At the metric-data level such a square has one constant coefficient and one nonconstant coefficient; it lies in the intermediate metric regime mentioned in [12, Example 8.20, p. 598]. This observation does not exclude specializing the book's generic square formulas to subfamilies, or more general extension constructions. The quotient and ordinary-product obstructions concern the specified coordinate maps.

Proposition 4.1 (Balanced zero-translation tensor block)**.** *On the declared real coefficient scheme,* $\Pi_{t,\pm} = (I \pm P_t)/2$ *are complementary idempotents of rank two. On the degree-zero tensor submodule generated by* $e^i \otimes e^i$*, the actual bimodule map has matrix*

$$\sigma_{t,0} = \mathscr{Q}_t^{\mathsf{T}} = -H_t P_t H_t^{-1}.$$

It satisfies $\sigma_{t,0}^2 = I$*,* $\operatorname{tr}\sigma_{t,0} = 0$ *and* $\det\sigma_{t,0} = 1$*, and its two eigensummands have rank two.*

Proof. The block equations give the complementary idempotents. Their traces are two, so their images are finite projective modules of rank two: over a residue field an idempotent has integer rank between zero and four, and equality of that rank with its trace two permits only rank two when 2 is invertible. This proves the rank statement even over nonreduced test algebras. The component formula for σ gives the degree-zero matrix $\mathscr{Q}_t^{\mathsf{T}}$. Transposing $\mathscr{Q}_t = -H_t^{-1}P_t^{\mathsf{T}}H_t$ gives the stated conjugacy. The rank-two splitting then gives the trace and determinant assertions. □

This gives the trace restriction a direct spectral interpretation in a tensor block of the connection. It does not declare P_t itself to preserve the translation degrees of individual one-forms, or claim that either splitting is orthogonal for the metric.

The mathematical motivation is therefore the coexistence of minimal tree-versus-cycle spatial mixing, coupled temporal data, a balanced tensor block and reversible reconstruction of genuine QLCs. At the specified rational metric, the theorem supplies an exact example of multiple star-preserving QLCs for one nondegenerate metric. It provides a controlled family for subsequent connection-dependent computations. The worked real connection at the end of Section 7 provides a concrete instance of these coupled blocks and their active entries. No curvature calculation, action principle, continuum limit or physical uniqueness principle selecting this particular support has been established by the count.

Finally, marking must not be confused with a multiplicative symmetry argument. The permutation group S_3 produces three distinct *unoriented* path supports, since swapping its two endpoints stabilizes a path. There are six mixed support patterns only if the choice of which slice carries the path is included. This is a combinatorial statement, not a new count of QLCs. A permutation of spatial directions usually also permutes the anisotropic metric parameters, so it need not act on the same fixed-metric fibre. No factor of six, quotient count or enlarged-sector real count is inferred here.

5 The arithmetic control fibre and the generic degree

The generic degree will be proved from one finite arithmetic fibre, but a count of solutions in that fibre alone is insufficient. We must exclude the entire boundary of the *same family*, including limits with varying metric parameters, and check its relative Jacobian. This section gives those two additional arguments. All finite-field computations below are in characteristic $p_* = 65537$, not characteristic two.

5.1 A common arithmetic presentation and its proper closure

Set

$$R = \mathbb{Z}[q_1, q_2, q_3, (2q_1q_2q_3(q_1-1)(q_2-1)(q_3-1))^{-1}], \qquad b_* = (p_*, q_1-4, q_2-9, q_3-49) \in \operatorname{Spec} R.$$

The inverted polynomial has residue $962 \neq 0$. In this section the control metric is $q_* = (4, 9, 49)$ over $\mathbb{F}_{p_*}$; it is distinct from the rational characteristic-zero metric q^0 used later. Write

$$\alpha_i = q_i - (u_i-1)^2, \quad \beta_i = q_i + u_i - 1, \quad Q = q_1q_2q_3, \quad D_\beta = Q\beta_1\beta_2\beta_3,$$
$$\overline{A} = q_2\alpha_1u_2^2 + q_1\alpha_2u_1^2, \qquad \overline{B} = q_3\alpha_2u_3^2 + q_2\alpha_3u_2^2,$$

and put

$$D_A = q_1q_2u_1u_2, \quad D_B = q_2q_3u_2u_3, \quad D_{ij} = q_iq_j\beta_i\beta_j, \quad N_{ij} = q_i\alpha_i\beta_j^2 + q_j\alpha_j\beta_i^2.$$

The five graph equations are

$$E_A = D_A\xi - \overline{A}, \quad E_B = D_B\eta - \overline{B}, \quad E_{ij} = D_{ij}c_{ij} - N_{ij} \quad (ij = 12, 13, 23).$$

The remaining four graph equations are

$$H_p = q_2\xi\eta + 4\alpha_2 = 0, \tag{23}$$
$$L_p = Q\xi\eta m + \xi R_{23} + \eta R_{12} = 0, \tag{24}$$
$$T = Q(m^2-1) + q_2q_3\alpha_1 + q_1q_3\alpha_2 + q_1q_2\alpha_3 = 0, \tag{25}$$
$$H_{\mathrm{graph}} = -Q^4(H_0(z,c) + mH_1(z,c)) = 0. \tag{26}$$

Here $z_i = \alpha_i/q_i$, and

$$R_{23} = q_1q_3\alpha_2 + q_1q_2\alpha_3, \qquad R_{12} = q_2q_3\alpha_1 + q_1q_3\alpha_2.$$

Here H_0, H_1 are the explicit invariant polynomials of the preceding reconstruction, not additional unspecified equations. Thus all nine equations are integer polynomials. Denote their ideal by I.

Use the full 23-factor localization product s from the declared sector. Its factors are the nine u_i, α_i, β_i; the three $\overline{A}, \overline{B}, Q\sum_i \alpha_i/q_i$; the four path-entry numerators; and the seven triangle factors consisting of the cleared completion determinant and six active-entry numerators. Each is the polynomial obtained in the source reconstruction, including the stated unit normalizations. Let

$$G = \operatorname{Spec}((R[u_1, u_2, u_3, m, \xi, \eta, c_{12}, c_{13}, c_{23}]/I)_s).$$

No factor is dropped in the argument.

Put $h = u_1u_2u_3\beta_1\beta_2\beta_3$. On $D(h)$ all five graph denominators are units. Eliminating their unique sections identifies G scheme-theoretically with a localization X of a four-equation family in (u_1, u_2, u_3, m). Useful unnormalized source equations are

$$P_H = \overline{AB} + 4Q\alpha_2u_1u_2^2u_3, \tag{27}$$

$$P_L = \overline{AB}m + u_2u_3(q_3\alpha_2 + q_2\alpha_3)\overline{A} + u_1u_2(q_2\alpha_1 + q_1\alpha_2)\overline{B}, \tag{28}$$

$$P_{\mathrm{top}} = T, \tag{29}$$

$$\Phi = D_\beta^4\,\mathcal{H}\left(\frac{\alpha_1}{\beta_1}, \frac{\alpha_2}{\beta_2}, \frac{\alpha_3}{\beta_3}, \frac{\beta_1}{q_1}, \frac{\beta_2}{q_2}, \frac{\beta_3}{q_3}, m\right), \tag{30}$$

where $\mathcal{H}(X, Y, m) = H_0(X_iY_i, X_iY_j + X_jY_i) + mH_1(X_iY_i, X_iY_j + X_jY_i)$. Each displayed clearing multiplier is a unit on X. In particular, the support is not replaced by the support of a larger unsaturated compact system.

Take the affine scheme closure of X by saturating its four-equation ideal by the full localization product over R, and let

$$\overline{X} \subset \mathbb{P}^4_R, \qquad [W : U_1 : U_2 : U_3 : M],$$

be its projective scheme closure. Both closures are formed *before* specialization to b_*. The morphism $\overline{X} \to \operatorname{Spec} R$ is projective and finitely presented, hence proper. We prove

$$\overline{X}_{b_*} = X_{b_*}, \qquad X_{b_*} \text{ is geometrically reduced of length } 512, \tag{31}$$

and verify the actual relative Jacobian on this fibre.

5.2 Exact arithmetic certificates and their meaning

The supplement under `certificates/generic/` supplies the full polynomials, not numerical samples of their values. A sparse polynomial is a list of integer coefficient–exponent pairs with an explicit variable order; univariate residues are in ascending coefficient order. Arithmetic in $\mathbb{F}_{p_*}[t]/(E)$ always means polynomial reduction modulo the displayed E. It is not assumed that E is irreducible.

The verifier `audit/generic_certificate.py` reconstructs the equations from (23)–(30), computes the stated substitutions, divisions and determinants, and checks every identity below. In particular its acceptance is based on equalities of polynomial coefficients and explicit unit identities, not on stored truth values or file digests. The finite data used in the proof have the following roles:

Certificate	Exact mathematical content
Finite chart boundary	All 26 nontrivial u/β support words; four mixed Bézout identities, six single-β Bézout identities, three double-β unit ideals, and two endpoint quotient units.
Exceptional lines	Eight complete double-α lines; the global saturation identity and one univariate Bézout identity per line.
Finite projection	Exact pseudo-remainder and resultant identities; the squarefree projection polynomial of degree 642.
Projection exclusion	Every factor of its degree-130 complement, including all degree-drop and zero-slope branches.
Finite rational representation	A squarefree degree-512 polynomial, all coordinate residues, original equation residues and unit tests.
Proper boundary	All global source coefficients, continuous pole-cone faces, two global colon identities and six signed one-pole eliminations.
Graph Jacobian	All nine original graph equations, all 23 original factors, full 9×9 and 4×4 determinants and their inverses.

Finite polynomial arrays too long to print are part of these certificates. The verification rules and the exhaustiveness argument are given next; the arrays are witnesses for those rules, not replacements for them.

5.3 All finite graph-chart boundary supports

Let Γ be the closure of the graph of X in $\overline{X} \times_R (\mathbb{P}^1_R)^5$. Its projection to $\overline{X}$ is proper and surjective on supports: its closed image contains the dense open X. Thus every proposed finite source boundary point has a graph lift after a residue-field extension.

For a point in the closure of a finite-type open immersion over a Noetherian base, the DVR specialization lemma supplies a trait whose generic point lies in the open and whose closed point is the proposed point. We use this with arbitrary residue extensions of $\mathbb{F}_{p_*}$. The parameters of the trait reduce to q_* but may vary along it. All the nonzero residue constants used below are therefore units on the trait. This standard closure lemma is recorded in Stacks, Tag 0CM2.

At a finite source point u_i, m are integral. At the control point $\beta_i - u_i$ has residue $3, 8, 48$ for $i = 1, 2, 3$ respectively. Hence u_i and β_i cannot both vanish at one index. Use the letters U, B, N for $u_i = 0$, $\beta_i = 0$, and neither zero. The complement of $D(h)$ consists of exactly the $3^3 - 1 = 26$ words other than NNN.

First consider a vanishing endpoint u_1 with u_2, u_3 units. If $a = v(u_1) > 0$, then $\overline{A}$ is a unit, $v(\xi) = -a$, and (23) gives $v(\eta) = v(\alpha_2) + a \geq a$. The equation $E_B = 0$ forces $\overline{B} = 0$ in the residue field. Dividing (24) by ξ on the generic point gives

$$R_{23} = -Q\eta m - (\eta/\xi)R_{12}.$$

Both terms on the right have positive valuation, so $R_{23} = 0$ in the residue field as well. The reflected endpoint gives $\overline{A} = R_{12} = 0$. These statements allow arbitrarily unequal positive orders of the vanishing factors.

The seven words with U letters and no B letters are excluded as follows:

Word	Contradiction
UNN	$\overline{B} = R_{23} = 0$ and the left endpoint quotient unit below.
NNU	$\overline{A} = R_{12} = 0$ and the right endpoint quotient unit below.
NUN	Both path coordinates have poles, contradicting $H_p = 0$.
UNU	Both path coordinates have poles, contradicting $H_p = 0$.
UUN	In L_p, ηR_{12} is the unique pole; $R_{12}(0, 0, u_3) = 2891 \neq 0$.
NUU	The reflected unique-pole argument; $R_{23}(u_1, 0, 0) = 3296 \neq 0$.
UUU	The unique finite triangle section contradicts the two trace equations.

For the two endpoint quotient calculations, form

$$A_L = \mathbb{F}_{p_*}[u_2, u_3]/(\overline{B}, R_{23})|_{u_1=0}, \qquad A_R = \mathbb{F}_{p_*}[u_1, u_2]/(\overline{A}, R_{12})|_{u_3=0}.$$

Both have dimension eight. On the β-open chart the triangle sections are regular; substitute them into H_{graph} and eliminate m against T. After clearing only unit denominators, the obstruction classes g_L, g_R have invertible multiplication maps with determinants

$$\det M_{g_L} = 35553, \qquad \det M_{g_R} = 61639 \quad \text{in } \mathbb{F}_{p_*}.$$

The certificate supplies the two eight-element monomial bases, all reduced coefficients and inverse classes satisfying $g_L g_L^{-1} = g_R g_R^{-1} = 1$. A class with an inverse cannot vanish at a geometric point. This argument does not assume the quotient algebras are reduced. At UUU, the triangle sections are

$$(c_{12}, c_{13}, c_{23}) = (59/36, 339/196, 824/441).$$

After nonzero scalar normalizations the two trace equations are $4924 - 13357m = 0$ and $1764m^2 + 2855 = 0$. The first forces $m = 1070$, and the second has residue $18263 \neq 0$ there.

There are twelve mixed U/B words. The words BUN, NUB give the middle double-pole contradiction; UUB, UBU, BUU are the adjacent or separated U contradictions just proved. At UBB and BBU, respectively one path coordinate has a pole and the other has the finite nonzero residue -56 and -11; at BUB both have poles. These contradict H_p. The four remaining words reduce to endpoint necessary pairs:

Word	Necessary univariate pair	Resultant modulo p_*
UBN	$(\overline{B}, R_{23})\|_{u_1=0,u_2=-8}$	49478
UNB	$(\overline{B}, R_{23})\|_{u_1=0,u_3=-48}$	62173
BNU	$(\overline{A}, R_{12})\|_{u_3=0,u_1=-3}$	33484
NBU	$(\overline{A}, R_{12})\|_{u_3=0,u_2=-8}$	13379

For each pair (f, g), the supplied multipliers satisfy $af + bg = 1$. For example the UBN pair, with indeterminate $v = u_3$, is

$$f = -4104v^2 + 1152v + 27648, \qquad g = -36v^2 + 72v - 12384,$$

and its identity is

$$(30332v + 32123)f + (15613v + 26845)g = 1 \quad \text{in } \mathbb{F}_{p_*}[v].$$

All four identities are verified coefficientwise and therefore exclude points over every extension field.

Next suppose that at least two β_i vanish and no u_i vanishes. For each of the three pairs, substitute $u_i = 1 - q_i$ into the five path equations E_A, E_B, H_p, L_p, T. The resulting polynomials generate the unit ideal in the remaining u coordinate, m, ξ, η over $\mathbb{F}_{p_*}$. The certificate lists these five polynomials and the exact finite-field elimination; its reduced Gröbner basis is [1] in each case. The triple-β word lies in these pair loci, so all four multiple-B words are covered. No triangle coordinate is assumed finite in this calculation.

It remains to consider the three words with exactly one B and no U. If β_i vanishes, its two incident triangle sections have poles, and the third is integral. From the graph equations,

$$\beta_i c_{ij} = \frac{\alpha_i \beta_j}{q_j} + \frac{\alpha_j \beta_i^2}{q_i \beta_j}.$$

Here α_i has the nonzero residue $q_i(1 - q_i)$. The degree-four part of the triangle equation in the incident pair, up to a nonzero base unit, is $314(c_{ij}^2 - c_{ik}^2)^2$. Multiplying the global equation by β_i^4 and taking its residue therefore forces

$$(\beta_j/q_j)^2 = (\beta_k/q_k)^2, \qquad \beta_j/q_j = \varepsilon \beta_k/q_k, \quad \varepsilon \in \{1, -1\}.$$

Thus there are precisely six necessary signed branches. Substitute each linear relation into the path equation and the two trace equations and clear the already unit coordinate denominators. The resulting path polynomial f and division-free trace resultant g have degrees

Vanishing factor	β_1	β_2	β_3
Degrees $(\deg f, \deg g)$, for each sign	(6,14)	(4,10)	(6,14)

Each of the six pairs has a supplied identity $af + bg = 1$. In forming g no leading coefficient of the linear trace equation is inverted: if the two equations are $am + b = 0$ and $cm^2 + d = 0$, the identity

$$a^2(cm^2 + d) - c(am - b)(am + b) = cb^2 + a^2 d \tag{32}$$

proves the necessary equation even where $a = 0$. The certificate reconstructs these six pairs from the original substituted equations before verifying their Bézout identities. All $7 + 12 + 4 + 3 = 26$ words are now excluded. Consequently the finite-source graph closure has no point on $V(h)$.

5.4 The eight exceptional lines on the remaining finite chart

On $D(h)$ the graph coordinates are uniquely reconstructed as schemes. However, one must not divide a resultant factor that can vanish there. At the control point write $a_i = \alpha_i$. On $D(h) \cap V(P_H)$,

$$V(a_2 \overline{B}) = V(a_1, a_2) \ \cup \ V(a_2, a_3). \tag{33}$$

Indeed, $\overline{B} = 0$ in (27) forces $a_2 = 0$, and then $P_H = q_2^2 u_2^4 a_1 a_3$. Conversely either pair makes P_H vanish. Thus the exceptional set is exactly the four lines

$$(u_1, u_2) \in \{-1, 3\} \times \{-2, 4\}$$

and the four lines

$$(u_2, u_3) \in \{-2, 4\} \times \{-6, 8\}.$$

Their intersections are retained. On the complement of (33), $a_2, \overline{A}, \overline{B}$ are all units.

To exclude each *whole* line in the global closure we need an equation in the global saturated ideal, not saturation after specialization. Define

$$C = 4u_2^2 m + 4u_2(u_1 + u_3) + \xi(u_3^2 - u_2^2) + \eta(u_1^2 - u_2^2).$$

Direct expansion over the integer polynomial ring gives

$$\begin{aligned} Q\xi\eta C - 4u_2^2 L_p &= 4q_3\eta E_A + 4q_1\xi E_B \\ &\quad + q_1 q_3[\eta(u_1^2 - u_2^2) + \xi(u_3^2 - u_2^2)]H_p =: W_0, \qquad (34) \\ D_B\eta E_A + \overline{A}E_B &= D_A D_B \xi\eta - \overline{AB}. \qquad (35) \end{aligned}$$

Combining them yields

$$Q\overline{AB}C = D_A D_B(4u_2^2 L_p + W_0) - QC(D_B\eta E_A + \overline{A}E_B) \in I.$$

Since $Q, \overline{A}, \overline{B}$ are inverted on G, this proves $C \in I : s^\infty$ over R. Every point of the specialized global closure must satisfy its specialization. No saturation–specialization interchange has been used.

On each of the eight lines, $4u_2^2$ is a unit. Substitute the five graph sections into $C = 0$ and solve uniquely for $m = M(v)$, with v the free coordinate. Substitute this expression into T and the actual triangle equation. Cancellation leaves denominators consisting of nonzero constants and powers of v, which are units on $D(h)$. The numerator pairs have the following degrees:

Fixed coordinates	Free coordinate	Top numerator	Triangle numerator
$(u_1, u_2) = (-1, -2)$	u_3	4	9
$(u_1, u_2) = (-1, 4)$	u_3	4	9
$(u_1, u_2) = (3, -2)$	u_3	4	9
$(u_1, u_2) = (3, 4)$	u_3	4	9
$(u_2, u_3) = (-2, -6)$	u_1	3	9
$(u_2, u_3) = (-2, 8)$	u_1	4	9
$(u_2, u_3) = (4, -6)$	u_1	4	9
$(u_2, u_3) = (4, 8)$	u_1	3	9

For example the first line gives $M(v) = (199v^2 - 202v - 144)/(392v)$. The complete numerator polynomials and multipliers $A(v), B(v)$ are supplied for all eight lines; the verifier reconstructs them from (34) and checks

$$A(v)T_C(v) + B(v)H_C(v) = 1 \quad \text{in } \mathbb{F}_{p_*}[v].$$

Thus every exceptional line, including every intersection, is empty in the specialized global closure.

5.5 An exhaustive finite projection on the allowed complement

For this subsection only, let $P_H^c, P_0^c, P_\triangle^c$ denote the three primitive compact control polynomials supplied in `certificates/generic/compact/`. They are defined from the four source equations by eliminating the scalar m without division. More explicitly, write the normalized source equations as

$$L = A_m m + B_m, \qquad \Phi_0 = C_m m + D_m, \qquad T_0 = T_2 m^2 + T_{00}.$$

The defining polynomial identities are

$$P_H = 2P_H^c,$$
$$A_m\Phi_0 - C_mL = A_mD_m - C_mB_m = 2P_\triangle^c, \tag{36}$$
$$A_m^2T_0 - T_2(A_mm - B_m)L = T_2B_m^2 + T_{00}A_m^2 = -4P_0^c. \tag{37}$$

For these compact identities the normalized polynomials are $L = P_L|_{q_*}$, $T_0 = -P_{\mathrm{top}}|_{q_*}$, and $\Phi_0 = \Phi|_{q_*}$; the path polynomial is $P_H^c = P_H|_{q_*}/2$. The nine-variable graph triangle equation, after substituting its sections and multiplying by D_β^4, equals $-Q^4\Phi$; here $Q = 1764$ and $-Q^4 = -9682651996416 \neq 0 \pmod{p_*}$. The verifier checks all these normalizations against the displayed graph equations, not merely against the compact data. They give necessary compact equations even where $A_m = 0$.

Work on $D(h\alpha_2\overline{B})$. In the variable u_1, the exact resultant is

$$\mathrm{Res}_{u_1}(P_H^c, P_0^c) = 13108\,u_2^8\alpha_2^4\overline{B}^2F(u_2,u_3) \quad \text{in } \mathbb{F}_{p_*}[u_2,u_3]. \tag{38}$$

Only the displayed units are cancelled. The last nonconstant subresultant is $\ell_1u_1 + \ell_0$; coefficient comparison also proves that it is the negative pseudo-remainder of P_0^c by P_H^c. Thus every common compact root satisfies it, including specialized leading-coefficient degeneracies. Define the polynomial

$$C_{\mathrm{proj}} = \sum_{i=0}^{10}[u_1^i]P_\triangle^c(-\ell_0)^i\ell_1^{10-i}.$$

If $\ell_1 \neq 0$, this is the denominator-cleared substitution $\ell_1^{10}P_\triangle^c(-\ell_0/\ell_1, u_2, u_3)$. If $\ell_1 = 0$, the necessary linear relation gives $\ell_0 = 0$; the same polynomial still vanishes. Hence every allowed compact point projects to $F = C_{\mathrm{proj}} = 0$.

The exact resultant of these two polynomials has u_3-degree 8568. Its monic squarefree part factors as

$$\mathrm{rad}\,\mathrm{Res}_{u_2}(F, C_{\mathrm{proj}}) = E(u_3)J(u_3), \qquad \deg E = 512, \quad \deg J = 130, \quad \gcd(E,J) = 1. \tag{39}$$

This is a coefficient identity over $\mathbb{F}_{p_*}$, and E, J are both supplied coefficientwise. The equality is used as a necessary projection cover, not as an assertion that all its roots lift.

Here is the complete exclusion rule for the complementary factor J. Factor J into its 37 irreducible factors j. For each, work in the finite field $k = \mathbb{F}_{p_*}[z]/(j)$ and compute in $k[y]$

$$G_y = \mathrm{rad}\,\gcd(F, C_{\mathrm{proj}}), \qquad D_y = \gcd(G_y, \ell_1), \qquad L_y = G_y/D_y.$$

These split every projected root into the zero-slope and nonzero-slope charts. For linear j, factor the resulting polynomials over $\mathbb{F}_{p_*}$; for the other factors the resulting nonconstant polynomials in y are linear, as checked by the certificate. In each resulting residue field, substitute y, z into *all three* compact polynomials and form the radical common polynomial in u_1. If they all vanish identically, the certificate checks that one of $u_2, u_3, \alpha_2, \beta_2, \beta_3$ vanishes identically, so the branch is outside the allowed complement. Otherwise remove its common roots with

$$u_1, u_2, u_3, \alpha_2, \beta_1, \beta_2, \beta_3$$

by repeated polynomial gcd and exact division. The remainder is a nonzero constant in every case. Thus there is no allowed J-root. The weighted degrees of the zero-slope and nonzero-slope projection parts are 129 and 4 respectively. They count projected (y, z) branches, so several y roots can lie over one root of J; they need not sum to $\deg J$ and are not counts of admissible source points. The certificate gives the full 37-factor inventory and every resulting branch, so neither a vanished slope nor an identically zero compact specialization is silently discarded.

For the E part, form the generally non-field algebra

$$A_* = \mathbb{F}_{p_*}[t]/(E(t)).$$

All Euclidean elimination pivots used below are certified units in A_*. The monic common divisor of $F(y,t), C_{\mathrm{proj}}(y,t)$ is linear in y, giving a unique $u_2 = r_2(t)$. The monic common divisor of $P_H^c(u_1, r_2, t), P_0^c(u_1, r_2, t)$ is linear in u_1, giving a unique $u_1 = r_1(t)$. Finally $A_m(r_1, r_2, t)$ is a unit and gives $m = r_m(t) = -B_m/A_m$. The three degree-below-512 residue arrays are included in the certificate. Direct substitution proves every compact and original source equation. Moreover,

$$\deg E = 512, \quad \gcd(E, E') = 1,$$

and all 23 original localization factors evaluate to units in A_*. For each factor f, this is checked as a polynomial inverse identity $f(t)v_f(t) \equiv 1 \pmod{E}$, not as a nonzero value at one sample. The source Jacobian is a unit as well, as detailed below.

The two linear gcd identities prove uniqueness of every allowed lift above each root of E. Conversely the explicit residues provide such a lift for every root. Thus exactly 512 geometric points remain. The eight exceptional lines and all 26 chart supports have already been excluded. Since $I : s^\infty$ and I agree after localization at s, the reconstructed points lie on the specialized global closure with the same local scheme structure. The Jacobian unit proves that these local rings are reduced, not merely that their support has size 512.

5.6 Genuine source infinity, including moving-base limits

Suppose a proposed point of $\overline{X}_{b_*}$ has $W = 0$. Apply the DVR closure lemma as above. At least one of u_1, u_2, u_3, m has a pole: otherwise $[1 : u_1 : u_2 : u_3 : m]$ extends in the affine chart, and separatedness of projective space forbids a different projective limit. Set

$$r_i = \max(0, -v(u_i)), \qquad r_m = \max(0, -v(m)).$$

The four global polynomials (27)–(30) have respectively 23,55,8,1198 monomials after grouping by exponents in (u_1, u_2, u_3, m). Every grouped coefficient specializes to a nonzero element at b_*. This is checked for every coefficient, including all 164616 integer monomials in the global expansion of Φ. Consequently these coefficients are units on any moving-base trait; the following initial-face arguments do not require the parameters to be constant along that trait. We express the control-field faces in the scalar normalization $P_H^c = P_H|_{q_*}/2$, $T_0 = -P_{\mathrm{top}}|_{q_*}$ and $\Phi_0 = \Phi|_{q_*}$; all these changes are by nonzero control-field scalars. The roles and normalizations are:

Role	Control polynomial / checker key	Centre numerator / file
Path	$P_H\|_{q_*}/2 = P_H^c$ / `PH4`	P_p / `Hp.txt`
Scalar path	$P_L\|_{q_*} = L$ / `PL4`	eliminated by $m = m_0$
Top	$-P_{\mathrm{top}}\|_{q_*} = T_0$ / `FT4`	P_F / `F.txt`
Triangle	$\Phi\|_{q_*} = \Phi_0$ / `PTH4`	P_T / `Htriangle.txt`

The control column is over $\mathbb{F}_{p_*}$ at q_*; the centre column is over $\mathbb{Q}$ at q^0 after scalar elimination. It records the roles of different polynomial arrays. The exact centre multipliers are (42)–(44). In particular the generic top equation is denoted P_{top}, and P_T denotes the centre triangle numerator.

The leading part of the top equation, up to a nonzero scalar, is

$$441u_1^2 + 196u_2^2 + 36u_3^2 - 1764m^2.$$

The largest positive pole order in this equation must occur at least twice. In particular $r_m \le \max_i r_i$, and m cannot be the only pole. If all three u_i have positive pole order, the path equation has the unique dominant monomial $377u_1^2u_2^4u_3^2$, a contradiction.

For exactly two positive u-pole orders, the following componentwise maximal faces apply throughout the *entire* positive rate cone:

Pole coordinates	Path factor in the finite coordinate	Triangle factor
u_2, u_3	$13u_1^2 - 18u_1 - 27$	$(u_1 + 3)^4$
u_1, u_3	$(13u_2^2 - 8u_2 - 32)(29u_2^2 - 49u_2 - 196)$	$(u_2 + 8)^4$
u_1, u_2	$29u_3^2 - 9u_3 - 216$	$(u_3 + 48)^4$

Only nonzero scalar factors and powers of the pole coordinates have been suppressed. Every other m-free monomial has coordinatewise exponents no larger than the displayed maximal pair and at least one strict inequality. Thus the same face applies throughout the positive rate cone. The triangle face forces the finite residue to be $-3, -8, -48$ respectively, where the displayed path factors are 144, $864 \cdot 2052$, and 67032, all nonzero modulo p_*. There are 501 m-linear triangle monomials. Their exponents in either positive coordinate are at most seven, while the m-free maximal face has exponents $(8, 8)$. Their weighted deficit is at least

$$r_i + r_j - r_m \geq \min(r_i, r_j) > 0.$$

Thus arbitrary unequal rates, the equality wall, and an additional m pole cannot change this face. Finite coordinates with positive valuation, and hence zero residue, are included by the displayed polynomial conditions.

Finally suppose exactly one u_i has positive pole order. The top equation forces $r_m = r_i$ and, at the leading coefficient level,

$$m/u_1 = \pm 1/2, \qquad m/u_2 = \pm 1/3, \qquad m/u_3 = \pm 1/7.$$

For any global polynomial $f = \sum_{\alpha,\ell} a_{\alpha,\ell}(q) u^\alpha m^\ell$ occurring here, define the one-pole chart polynomial by

$$d_i(f) = \max_{a_{\alpha,\ell} \neq 0} (\alpha_i + \ell), \qquad f^{\infty,i}(\widehat{u}_i, M) = t^{d_i(f)} f(u_i = t^{-1}, m = M/t, \widehat{u}_i; q)\Big|_{t=0,\, q=q_*}.$$

Here $\widehat{u}_i$ denotes the two remaining u coordinates. All grouped coefficients are units on the moving-base trait, so the same weighted support applies before and after reduction. This formula is polynomial in the finite coordinates, including zero residues, and defines $\Phi^{\infty,i}$ in particular.

For an endpoint $i = 1$ or 3, put $t = 1/u_i$ and $M = m/u_i$. The other source coordinates and M are integral. Let

$$F_H = t^2 P_H, \qquad F_L = t^3 P_L, \qquad E_1 = \overline{B}, \quad E_3 = \overline{A},$$
$$D_i = q_i \alpha_2 - q_2 u_2^2, \qquad N_i = M D_i - q_2 u_2.$$

At $t = 0$ the two chart equations give $E_i D_i = E_i N_i = 0$. If $E_i \neq 0$, then $D_i = N_i = 0$ forces $u_2 = 0$ and then $D_i = q_i(q_2 - 1) \neq 0$. Therefore $E_i = 0$.

Exact division over $\mathbb{Z}[q_1, q_2, q_3]$ gives a 269-term polynomial K_i for each endpoint satisfying

$$t u_2 K_i = D_i F_L - N_i F_H. \tag{40}$$

On the trait's generic point t and u_2 are nonzero, so $K_i = 0$ there and then in the valuation ring. This uses neither division by the residue of u_2 nor saturation after specializing $t = 0$. For each sign of M, eliminate one finite coordinate from

$$E_i = 0, \quad K_i|_{t=0} = 0, \quad \Phi^{\infty,i} = 0.$$

The exact pairs of equations and the eliminated coordinate are:

i	Two pairs	Eliminate	Retain
1	$(E_1, K_1\|_{t=0}),\ (E_1, \Phi^{\infty,1})$	u_2	u_3
2	$(P_H^{\infty,2}/2, P_L^{\infty,2}),\ (P_H^{\infty,2}/2, \Phi^{\infty,2})$	u_1	u_3
3	$(E_3, K_3\|_{t=0}),\ (E_3, \Phi^{\infty,3})$	u_1	u_2

All entries are reduced at q_* and at each signed value of M above; the factor $1/2$ is the control path normalization. The endpoint resultant degrees are 18 and 32, and the middle-chart degrees are 8 and 32. For all six signed cases the certificate supplies their complete coefficient arrays A, B and multipliers U, V satisfying

$$UA + VB = 1 \quad \text{in } \mathbb{F}_{p_*}[z].$$

The verifier first reconstructs each resultant from the actual chart polynomials by finite-field Sylvester determinants and interpolation under a checked degree bound, and only then checks the unit identity. A common chart root makes both resultants vanish even if a specialized leading coefficient drops: this is the polynomial Sylvester identity, not division by that leading coefficient. Each signed chart is therefore empty over the algebraic closure.

The cases of zero, one, two and three positive u-pole orders exhaust every possible pole vector. We have proved

$$\overline{X}_{b_*} \cap V(W) = \varnothing. \tag{41}$$

5.7 The actual relative Jacobian and the finite étale neighbourhood

In the algebra $A_* = \mathbb{F}_{p_*}[t]/E$, reconstruct all nine coordinates by the preceding rational representation and the five graph sections. Differentiate *all nine original graph equations* with respect to their nine variables, substitute these residues and compute the determinant by a division-free subset recurrence. Explicitly, for a matrix J over A_*, set $d_\varnothing = 1$ and, for a column subset S of size k,

$$d_S = \sum_{j \in S} (-1)^{\#\{\ell \in S : \ell > j\}} J_{k,j} d_{S \setminus \{j\}}.$$

Then $d_{\{1,\dots,9\}} = \det J$. This uses no pivot divisions and therefore remains valid in the generally reducible algebra A_*. The resulting determinant has a supplied polynomial inverse modulo E. Independently, differentiate the four actual source equations in their four variables. Their determinant is also a unit, and elimination of the five graph-section rows verifies the Schur-complement equality with the nine-variable determinant. The coefficients of every one of the 23 original factors are evaluated afresh in the same algebra and each product with its supplied inverse reduces to one. These tests certify the actual original open and Jacobian, rather than those of a convenient replacement system.

Proposition 5.1. *The whole projective control fibre is equal to the original localized source fibre. It is geometrically reduced of length 512, and the original source morphism is étale at each of its points.*

Proof. The proper graph projection reduces every finite source boundary point to the graph boundary. Section 5.3 excludes $V(h)$. On $D(h)$ the graph is a regular closed section; the eight-line argument and the exhaustive projection calculation then exclude all remaining localization boundaries and leave exactly the reconstructed 512 admissible points. Equation (41) excludes every point outside the finite source chart. Hence the entire support of $\overline{X}_{b_*}$ lies in X.

An open subscheme containing every point of a scheme is the entire scheme. Restriction of global scheme closure to an open commutes with the closure construction, and localization commutes with base change. Since $(I : s^\infty)_s = I_s$ before specialization, the control fibre has precisely the source scheme structure at these points; no nilpotent or embedded part is dropped. The source Jacobian units prove that all these local rings are reduced. They also prove relative étaleness by the square Jacobian criterion: there are nine equations in nine graph variables, or equivalently four equations in four source variables, with an invertible determinant. This criterion includes flatness; reducedness of a special fibre alone would not suffice. □

Theorem 5.2. *For the declared marked localized source family there is a nonempty base open $U \subset \operatorname{Spec} R$, containing b_*, such that*

$$\overline{X}_U = X_U \longrightarrow U \quad \textit{is finite étale of rank } 512.$$

In particular its characteristic-zero generic algebra over $\mathbb{Q}(q_1, q_2, q_3)$ has dimension 512 and is étale. After extension to an algebraic closure its generic fibre has exactly 512 reduced geometric points. The source is not required to be irreducible.

Proof. Let O be the étale locus of X, an open subscheme of $\overline{X}$. By Proposition 5.1, the entire control fibre lies in O. The closed complement $Z = \overline{X} \setminus O$ has closed image in the base because $\overline{X}$ is proper. This image misses b_*. Remove it and call the remaining nonempty open U. Then $\overline{X}_U = O_U = X_U$, so this morphism is both proper and étale. An étale morphism is locally quasi-finite; a proper locally quasi-finite morphism is finite. A finite étale algebra is finite locally free, with locally constant rank. The open U is irreducible because R is a domain, and its rank is 512 at b_*. It therefore has rank 512 everywhere on U, in particular at the characteristic-zero generic point. This proves the theorem. □

The standard scheme-theoretic ingredients used here are restriction of scheme closure to opens (Stacks, Tag 01R5), the square Jacobian criterion (Tag 02GU), proper and locally quasi-finite implies finite (Tag 02LS), and finite étale local freeness (Tag 02K9). The finite degree conclusion concerns this precise localization and marking; it asserts no unmarked quotient, full moduli classification or physical multiplicity.

6 A uniform length bound at every finite fibre

The generic degree computed in the preceding section is not by itself an upper bound at a special metric. The following argument supplies precisely the missing specialization statement. It applies to the original square presentation, not to a determinantal plane quotient.

Lemma 6.1 (Finite fibres of a square family)**.** *Let A be a Noetherian integral $\mathbb{Q}$-algebra, $K = \operatorname{Frac}(A)$, and*

$$B = A[x_1, \dots, x_n, 1/\Delta]/(f_1, \dots, f_n), \qquad f_i, \Delta \in A[x_1, \dots, x_n].$$

Suppose $\dim_K(B \otimes_A K) = d < \infty$. If the entire fibre over $s \in \operatorname{Spec} A$ is finite, then

$$\dim_{\kappa(s)}(B \otimes_A \kappa(s)) \leq d.$$

The statement includes all multiplicities and does not assume properness, reducedness of the special fibre, or that B is finite as an A-module.

Proof. The empty fibre satisfies the assertion. Otherwise write its finitely many support points as primes $\mathfrak{p}_1, \dots, \mathfrak{p}_k$ of B. They lie in the quasi-finite locus U of $\operatorname{Spec} B \to \operatorname{Spec} A$. This locus is open by the quasi-finite-locus theorem [4]. Write the complement as $V(J)$. For every i, $J \not\subset \mathfrak{p}_i$. Finite prime avoidance gives $h \in J$ outside all the $\mathfrak{p}_i$. Consequently $D(h) \subset U$ and $D(h)$ contains the whole special fibre. The image of h is outside every maximal ideal of its Artinian coordinate algebra; it is therefore a unit. Localizing at h preserves that entire fibre as a scheme, including its nilpotents.

Choose a polynomial representative H with $h = H/\Delta^j$ in B. Then, canonically,

$$B_h = A[x_1, \dots, x_n, 1/(\Delta H)]/(f_1, \dots, f_n).$$

Its morphism to $\operatorname{Spec} A$ is locally quasi-finite and has a localized square polynomial presentation. The square-presentation criterion for quasi-finite syntomic morphisms [5], condition (4) implying (2), says that this morphism is flat. This use of flatness does not assume the base regular. Introducing one inverse variable replaces the presentation by $n + 1$ variables and $n + 1$ equations and gives the same criterion.

The remaining argument bounds fibre dimension for a flat module that need not be finitely generated. Let $M = B_h$ as an A-module. Its generic fibre is a localization of the finite-dimensional algebra $B \otimes_A K$, so it has dimension at most d. No finiteness of M as an A-module has been inferred. Put $N = d + 1$. Since $N!$ is invertible in A, the antisymmetrizer

$$e = \frac{1}{N!} \sum_{\pi \in S_N} \operatorname{sgn}(\pi)\pi \quad \text{on } M^{\otimes_A N}$$

is an idempotent whose image is canonically $\bigwedge_A^N M$. Indeed, it annihilates repeated arguments and descends through the exterior quotient, whose composition with the descended map is the identity. Thus this exterior power is a direct summand of a flat module and is flat. Over the domain A it is consequently torsion-free. Its localization to K is zero, since $\dim_K(M \otimes_A K) \leq d$. A torsion-free module over a domain injects into its generic localization, so $\bigwedge_A^N M = 0$.

Exterior powers commute with base change. Therefore

$$\bigwedge_{\kappa(s)}^{N} (M \otimes_A \kappa(s)) = 0,$$

which forces the vector-space dimension to be at most $N - 1 = d$. The localization at h preserved the full original special fibre, so this is the required length bound. □

Corollary 6.2. *Every finite fibre of the declared localized connection family in characteristic zero has scheme length at most* 512*.*

Proof. After fixing the four invertible scales, use the integral metric base

$$A = \mathbb{Q}[q_1, q_2, q_3, (q_1 q_2 q_3 \prod_i (q_i - 1))^{-1}].$$

The scheme isomorphisms in Section 3 give four source variables (u_1, u_2, u_3, m) and the four rational equations

$$m^2 + S - 1 = 0, \quad AB + 4z_2 = 0, \quad ABm + A(z_2 + z_3) + B(z_1 + z_2) = 0, \quad H_0 + mH_1 = 0.$$

Here all quantities are the explicit functions defined in that section. Multiply each equation by its denominator and combine those denominators with the numerators of all inverted factors into a single Δ. On $D(\Delta)$ every multiplier is a unit. Hence this gives a square presentation of exactly the same scheme, not only the same field-valued points. Its generic algebra has dimension 512 by Theorem 5.2. Lemma 6.1, with $n = 4$, proves the assertion. The same argument after base extension to $\mathbb{C}$ gives the geometric-fibre statement. □

For clarity, a finite fibre has not yet been proved to have length *equal* to 512. For example, the square family $\mathbb{Q}[t, x]/(tx^2 + x)$ has generic length two and special length one at $t = 0$. The exact equality below therefore needs both finiteness and 512 distinct lifted points. Neither a generic degree nor a list of modular points is silently substituted for these two obligations.

7 The exact fibre at a rational metric

We now specialize to

$$q^0 = (4/3, 12/11, 3/4)$$

and keep the four scale parameters equal to one. Let C be the rational coordinate algebra of the original marked localized coefficient scheme at this metric. Thus every original open condition remains inverted; C is not the algebra of the unfiltered projection locus. We prove finiteness first, before applying the length bound of Lemma 6.1. This order prevents a generic degree from being used as an unjustified bound on an infinite exceptional fibre.

All large integer polynomials used below are supplied in full, with ascending univariate coefficient arrays and expanded multivariate integer coefficients, in `certificates/centre/`. The descriptions below define the exact finite identity tests; no numerical root approximation or prior review verdict is needed. In particular the two source minors, the eliminant, its two factors, the reconstruction cofactors and all modular coordinate residues are actual delivered polynomials.

7.1 A two-generator plane module proves finiteness

Use the scalar elimination and inverse reconstruction established in the source reduction, and write $y = u_2$, $z = u_3$, $v = 1/u_1$. Denote the three primitive integer numerator polynomials after eliminating m by

$$P_p, P_F, P_T \in \mathbb{Z}[u_1, y, z].$$

Their degrees in u_1 are $2, 6, 10$, respectively. In terms of the source quantities $\zeta_j = 1 - (u_j - 1)^2/q_j$, the eliminated scalar is

$$m_0 = \frac{A(\zeta_2 + \zeta_3) + B(\zeta_1 + \zeta_2)}{4\zeta_2}, \quad A = \zeta_1 u_2/u_1 + \zeta_2 u_1/u_2, \quad B = \zeta_2 u_3/u_2 + \zeta_3 u_2/u_3.$$

Here ζ_2 is already a unit. The files `Hp.txt`, `F.txt` and `Htriangle.txt` contain P_p, P_F, P_T, with the source meanings specified by that reduction.

Define

$$D = 27y^2z^2 - 32y^2z + 4y^2 - 22yz^2 - z^2,$$
$$J = 11y^2 - 22y - 1, \qquad K = 20y^2 - 22y - 1, \qquad E = 3D + 8zJ.$$

For clarity, the exact source-to-file identities at this metric are

$$H_p = \frac{P_p}{144u_1u_2^2u_3}, \tag{42}$$

$$m_0^2 + S - 1 = \frac{P_F}{2304u_1^2u_2^2u_3^2J^2}, \tag{43}$$

$$\mathcal{H}_0 + m_0\mathcal{H}_1 = \frac{P_T}{20736u_1u_2u_3J(4u_3 - 1)^4(11u_2 + 1)^4(3u_1 + 1)^4}. \tag{44}$$

These rational identities are checked by complete coefficient comparison in Section 3.8. Their denominators are products of declared units: $J = -11\alpha_2$, and the three linear factors are $4\beta_3, 11\beta_2, 3\beta_1$. In these identities m_0 is the displayed $C_p/(4\zeta_2)$, where $C_p = A(\zeta_2+\zeta_3)+B(\zeta_1+\zeta_2)$. The alternative $-C_p/(AB)$ agrees with it on $H_p = 0$; it is not substituted off that equation when deriving these numerator files. The identities $D = -12yzB$ and $J = -12\zeta_2$ show that every original point is on the open with coordinate ring

$$R = \mathbb{Q}[y, z, (yzDJ)^{-1}].$$

Reversal in the invertible variable gives

$$\phi = v^2P_p(1/v, y, z) = av^2 + bv + c, \qquad a = -3y^2D, \quad b = -6y^2E, \quad c = KD.$$

The leading coefficient a is a unit of R. Consequently

$$P = \phi/a = v^2 + \lambda v + \mu, \qquad \lambda = 2E/D, \quad \mu = -K/(3y^2)$$

defines a free rank-two R-algebra with basis $(1, v)$.

Put $U = v^6P_F(1/v, y, z)$ and $T_0 = v^{10}P_T(1/v, y, z)$. Full integer coefficient multiplication verifies

$$a^5U = Q_F\phi + 216y^6J^2(rv + s), \tag{45}$$

$$a^9T_0 = Q_T\phi + 1944y^{10}J(tv + w). \tag{46}$$

The quotients Q_F, Q_T and the entire remainders are delivered, so these are coefficient identities, not sampled equalities. The term counts of (r, s, t, w) are $(157, 173, 797, 816)$; their respective bidegrees in (y, z) are $(18, 14), (18, 14), (35, 27), (35, 27)$. All coefficients removed in (45)–(46) are units of R. Hence the ideals generated by (ϕ, U, T_0) and $(P, rv + s, tv + w)$ are equal in $R[v]$. This equality holds over arbitrary R-algebras, including nonreduced ones.

The candidate algebra

$$\mathcal{B} = R[v]/(P, rv + s, tv + w)$$

has the exact module presentation

$$R^4 \xrightarrow{\mathcal{K}} R^2 \longrightarrow \mathcal{B} \longrightarrow 0, \qquad \mathcal{K} = \begin{pmatrix} s & -\mu r & w & -\mu t \\ r & s - \lambda r & t & w - \lambda t \end{pmatrix}. \tag{47}$$

Indeed, the four columns are the coordinates of $(rv + s), v(rv + s), (tv + w), v(tv + w)$ in the basis $(1, v)$, and these generate the ideal as an R-module. No division by r or t is used here.

Multiply columns two and four by a, a unit, and write $\widehat{\mathcal{K}} = \mathcal{K}\,\mathrm{diag}(1, a, 1, a)$. Define $f, g \in \mathbb{Z}[y, z]$ by the exact normalization identities

$$\det \widehat{\mathcal{K}}_{12} = 12y^2 D^7 f, \qquad \det \widehat{\mathcal{K}}_{23} = 6y^2 D^9 g.$$

The full polynomials are `minor_12.txt` and `minor_23.txt`. The other four normalized minors are also supplied and checked; no factor unsupported on an existing unit is discarded. The relevant degrees are

$$\deg_{y,z} f = (24, 16), \quad \deg_{y,z} g = (37, 25), \qquad \#\,\mathrm{terms}(f) = 281, \quad \#\,\mathrm{terms}(g) = 833.$$

Lemma 7.1. *The polynomials f, g are coprime in $\mathbb{C}[y, z]$.*

Proof. Exact extraction gives $[y^0]f = -z^{16}$ and $([y^{24}]f)(0) = -40960000$. Thus f, considered in $\mathbb{C}[z][y]$, has no nonconstant content factor. Write $F_1(y) = f(y, 1)$ and $G_1(y) = g(y, 1)$. Their degrees are 24 and 37; their leading coefficients modulo 101 are 4 and 83, respectively. An explicit extended-Euclidean certificate, recomputed from these coefficient arrays, gives

$$a_* \overline{F}_1 + b_* \overline{G}_1 = 1 \quad \text{in } \mathbb{F}_{101}[y], \qquad \deg a_* = 36, \quad \deg b_* = 23.$$

As a separate test the complete ascending-coefficient 61×61 Sylvester matrix has determinant 98 modulo 101.

If a primitive nonconstant integer polynomial divided F_1, G_1, its reduction would still have positive degree: reduction preserves the entire degree of F_1, so it preserves the degree of each factor of F_1. This contradicts the displayed identity. Thus F_1, G_1 are coprime over $\mathbb{Q}$, and a rational Bézout identity proves the same over $\mathbb{C}$. A common factor of f, g of positive y-degree would stay nonconstant at $z = 1$, because that specialization preserves the full y-degree of f: the leading y-coefficients of factors multiply, and their product is nonzero at $z = 1$. It would divide F_1, G_1. A common factor of y-degree zero would divide the content of f. Both possibilities have been excluded. □

Theorem 7.2. *The entire original localized fibre at q^0 is finite, including every possible nonreduced contribution.*

Proof. Two coprime polynomials in two variables have a zero-dimensional common zero scheme: a positive-dimensional component would give a height-one prime, which is principal in the polynomial ring, and hence a common factor. Therefore $\mathbb{Q}[y, z]/(f, g)$, and its localization $R/(f, g)$, are finite-dimensional vector spaces.

Every maximal minor of a two-row presentation matrix annihilates its cokernel. For a selected pair of columns, their 2×2 adjugate expresses the determinant times either basis vector in the image. The two displayed minors differ from f, g by units, so $(f, g)\mathcal{B} = 0$. Presentation (47) gives a surjection $(R/(f, g))^2 \twoheadrightarrow \mathcal{B}$. Thus $\mathcal{B}$ is finite dimensional. The original coordinate algebra C is obtained by the inverse change of variables, the unique source reconstruction and all remaining original localizations. Localization of a finite-dimensional algebra selects Artinian factors and remains finite dimensional. These are equalities of localized schemes; in particular the argument preserves nilpotents. No rank-zero stratum or additional localization boundary was discarded. □

Applying Lemma 6.1 to the original square family, with generic degree 512 and the finite fibre just proved, now gives

$$\dim_{\mathbb{Q}} C = \dim_{\mathbb{C}}(C \otimes \mathbb{C}) \leq 512. \tag{48}$$

The equality follows from extension of scalars for a finite vector space. This is the first point in the argument at which the generic degree is used to bound this particular fibre.

7.2 A complete smooth residue witness and its lifts

The file `candidate_mod101.json` supplies eleven monic polynomials $h_i(Z)$ and residues U_{1i}, U_{2i}, U_{3i} in

$$T_i = \mathbb{F}_{101}[Z]/(h_i), \qquad U_{3i} = Z \pmod{h_i}.$$

The certificate is fully checked in these quotient rings by polynomial convolution, Euclidean division and explicit inverses. The degrees are

$$(1, 1, 2, 2, 5, 6, 56, 61, 62, 158, 158), \qquad \sum_i \deg h_i = 512. \tag{49}$$

The tests are

$$\gcd(h_i, h_i') = 1, \qquad \gcd(h_i, h_j) = 1 \quad (i \neq j), \tag{50}$$

$$P_p(U_i) = P_F(U_i) = P_T(U_i) = 0, \qquad \det \frac{\partial(P_p, P_F, P_T)}{\partial(u_1, u_2, u_3)}(U_i) \in T_i^{\times}. \tag{51}$$

The Jacobian is that of the original three numerator equations, rather than of the two convenient projection minors. The first two components are $h_1 = Z + 48$, $U_1 = (29, 82, 53)$ and $h_2 = Z + 67$, $U_2 = (47, 39, 34)$; the remaining, possibly reducible, component arrays are supplied in full. Irreducibility is unnecessary: (50) gives 512 distinct geometric values of u_3.

Every reconstruction denominator is a unit in each T_i. In particular, with

$$\alpha_j = q_j - (u_j - 1)^2, \quad \beta_j = q_j + u_j - 1, \quad d_j = q_j u_j/\beta_j, \quad X_j = \alpha_j/\beta_j, \quad Y_j = \beta_j/q_j,$$

the full list of original factors checked is

$$\begin{gathered} d_j,\ q_j - d_j,\ q_j - d_j^2 \quad (j = 1, 2, 3), \\ S, A, B, E_1, p_0, 2 - p_0, t_0, 2 - t_0, p, 2 - p, r_1, 2 - r_1, t_1, 2 - t_1, \end{gathered} \tag{52}$$

where $S = \sum_j \zeta_j$, $E_1 = \sum_j X_j^2(Y_{j+1}^2 - Y_{j+2}^2)$ with cyclic indices, $p_0 = 2\zeta_2 u_1/(u_2 A)$ and $t_0 = 2\zeta_3 u_2/(u_3 B)$. A unit test means a polynomial gcd equal to one, equivalently an inverse identity modulo the entire h_i. The triangle is reconstructed from its three right-kernel equations, two left-kernel equations and its trace equation; the determinant is $Y_3 E_1$. The remaining left-kernel equation and all ten original coefficient residuals are checked after reconstruction. The linear remainder coefficient $r(U_{2i}, U_{3i})$ is separately a unit.

Lemma 7.3. *The original characteristic-zero fibre has 512 distinct points after extension to a single finite extension of $\mathbb{Q}_{101}$, with 512 distinct values of u_3.*

Proof. Choose a finite field $k/\mathbb{F}_{101}$ splitting all h_i, and let $\mathcal{O}$ be the complete ring of integers of the corresponding finite unramified extension $L/\mathbb{Q}_{101}$. Each residue tuple is a common zero of the three integer source polynomials with invertible Jacobian. Suppose $a_k \in \mathcal{O}^3$ solves these equations modulo 101^k. Choose a correction $t \pmod{101}$ solving

$$J(a_k)t \equiv -101^{-k}(P_p, P_F, P_T)(a_k) \pmod{101}, \qquad a_{k+1} = a_k + 101^k t.$$

The invertible residue Jacobian makes this possible, and the polynomial Taylor formula proves the equations modulo 101^{k+1}. Completeness gives an exact source zero in $\mathcal{O}^3$.

All required rational constants are defined at 101. Every denominator and factor in (52) remains a unit, since its residue was checked nonzero. Source reconstruction therefore gives an L-point of the original localized fibre. Distinct residue tuples have distinct lifts. More strongly, their u_3 residues are all distinct by (50), so their lifted u_3 values are distinct. Only this witnessed subset is lifted; no assertion that the entire special fibre has been described is needed. □

Theorem 7.4. *The original marked localized coefficient fibre at q^0 has exactly 512 distinct reduced nonsingular complex points and scheme length 512:*

$$C \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C}^{512}.$$

Proof. The 512 L-points of Lemma 7.3 give 512 distinct maximal ideals with residue field L in $C \otimes L$. The Chinese remainder theorem gives a surjection

$$C \otimes L \twoheadrightarrow L^{512}.$$

The source has dimension at most 512 by (48), so the surjection is an isomorphism and $\dim_{\mathbb{Q}} C = 512$. An additional point or a nonzero nilpotent contribution would add positive length, so both are excluded by this equality. Faithful field extension implies that C is reduced. A finite reduced rational algebra is a product of finite separable number fields; after extension to $\mathbb{C}$ these split into exactly 512 copies of $\mathbb{C}$. The original three-source Jacobian is nonzero at every lift by its nonzero residue, so its determinant is a unit in $C \otimes L$ and in C. The source equivalence preserves this nonsingularity. This argument does not identify $\overline{\mathbb{Q}}_{101}$ with $\mathbb{C}$, assume properness, or deduce characteristic-zero equality from a modular count alone. □

Corollary 7.5. *The coordinate u_3 generates C as a rational algebra, and the coefficient $r(u_2, u_3)$ is a unit in C.*

Proof. In $C \otimes L = L^{512}$, the 512 entries of u_3 are distinct. Vandermonde interpolation gives $L[u_3] = L^{512}$. The inclusion $\mathbb{Q}[u_3] \subset C$ becomes an equality after faithful extension to L, hence was an equality already. At every lift, r has nonzero residue. Thus multiplication by r is invertible on $C \otimes L$; its determinant is then nonzero over $\mathbb{Q}$, so r is a unit in C. The relation $rv + s = 0$ gives $v = -s/r$ everywhere, including every point of the original fibre. □

7.3 Identification of the explicit rational univariate algebra

The two given source minors define the integer resultant

$$R_*(z) = \operatorname{Res}_y(f(y,z), g(y,z)), \qquad \deg R_* \le 24 \cdot 25 + 37 \cdot 16 = 1192.$$

The exact supplied coefficient arrays satisfy

$$R_* = c_* H E_*, \quad c_* \in \mathbb{Z} \setminus \{0\}, \quad \deg R_* = 1192, \quad \deg H = 512, \quad \deg E_* = 680, \tag{53}$$

with H, E_* primitive. This complete product identity is checked by integer convolution. The source resultant identity is checked by a separate coefficient parser and exact integer univariate resultants at 1193 distinct negative integers. At every test the y-degrees remain 24 and 37. Both the actual resultant and the proposed array have degree at most 1192, so these tests prove equality of polynomials over $\mathbb{Z}$. These are a complete identity proof, rather than random spot checks. FLINT supplies the exact integer resultant backend; the product and modular factor tests use only Python integers.

The complete complementary factor, with every multiplicity retained, satisfies

$$\operatorname{lc}(H) \equiv 10 \pmod{101}, \quad \overline{H}/\overline{\operatorname{lc}(H)} = \prod_i h_i, \qquad \gcd(\overline{E}_*, h_i) = 1 \quad \text{for every } i. \tag{54}$$

Theorem 7.6. *The supplied polynomial H, after monic normalization, is the minimal polynomial of u_3 in the original rational coordinate algebra:*

$$C = \mathbb{Q}[u_3] \cong \mathbb{Q}[Z]/(H(Z)).$$

In particular this algebra is identified in characteristic zero, not merely by agreement of its reduction modulo 101.

Proof. At each of the 512 integral lifts, the source minors f, g vanish. The Sylvester adjugate identity therefore gives $R_*(u_3) = 0$, even if a leading coefficient of f or g drops at that point. By (54), $E_*(u_3)$ has nonzero residue and is a unit in $\mathcal{O}$. The exact identity (53) and $c_* \neq 0$ in the characteristic-zero field L imply $H(u_3) = 0$. The scalar c_* need not be a 101-adic unit; it is cancelled in L. Thus H annihilates u_3 in $C \otimes L$ and, by faithful base change, in C. Corollary 7.5 shows that its minimal polynomial has degree 512. The displayed H has the same degree, so its monic normalization is exactly that polynomial. □

7.4 A complete exact real-root certificate

Write $H(x) = \sum_{i=0}^{512} a_i x^i$. The coefficient arrays satisfy

$$\sum_{i<512} |a_i| B^i < |a_{512}| B^{512}, \qquad B = 2^{19}.$$

For $|x| \geq B$, each ratio $|a_i x^i|/|a_{512} x^{512}|$ is no larger than at B. Thus all complex roots have modulus strictly less than B. In particular every real root is in $(-B, B)$.

Parameterize the negative and positive halves by $x = -Bt$ and $x = Bt$, $0 < t < 1$. For a node polynomial $Q(t)$ of degree n, define

$$\mathcal{D}Q(s) = (1+s)^n Q(1/(1+s)).$$

For $s > 0$, this change of variables bijects to $0 < t < 1$; its prefactor never vanishes there. Descartes' rule of signs bounds the positive-root multiplicity count by the variation V of the nonzero coefficients of $\mathcal{D}Q$, and their difference is even. In particular $V = 0$ proves emptiness and $V = 1$ proves exactly one root. Coefficient reversal followed by a Taylor shift by one computes $\mathcal{D}Q$ exactly.

For $V > 1$ bisect, using the integer coefficient transformations

$$Q_0(t) = 2^n Q(t/2), \qquad Q_1(t) = Q_0(t+1).$$

Positive integer content can be removed without changing roots or signs. A root at a subdivision midpoint is tested exactly and would be counted once, separately from the two open children. The delivered finite trees, in `descartes_partition.json`, have the following independently checked data:

Half-axis	One-root leaves	All leaves	Nodes	Maximum depth
$u_3 < 0$	46	85	169	38
$u_3 > 0$	158	233	465	35
Total	204	318	634	

Every leaf is checked to have variation zero or one. The leaf paths are distinct and prefix-free, and every internal prefix has both children. Therefore the finite trees cover the initial intervals, apart from the separately tested splitting points. None of the splitting points is a root, and $H(0) \neq 0$. A second integer-only checker independently verifies that all 319 distinct partition endpoints are nonroots and that each of the 204 one-root leaves has opposite endpoint signs. This also supplies a direct intermediate-value lower bound. No assumed termination of an infinite subdivision is used: the delivered finite tree is the complete certificate.

Theorem 7.7. *The original localized real coefficient fibre has exactly 204 points:*

$$N_{\mathbb{R}} = 204, \qquad N_{\mathbb{R},u_3<0} = 46, \quad N_{\mathbb{R},u_3>0} = 158.$$

The remaining 308 points of the complexification form 154 conjugate pairs. The 204 real coefficient points give the real, star-preserving connections in the declared sector through the source equivalence.

Proof. The strict root bound, full binary coverage, endpoint tests and Descartes variations prove that H has exactly $46 + 158 = 204$ real roots. By Theorem 7.6, real points are precisely the rational algebra homomorphisms $C \to \mathbb{R}$, which are exactly evaluations at those roots. Every original coordinate and every localized inverse belongs to $C = \mathbb{Q}[u_3]$, so it evaluates to a real value; no additional reality filter is missing. The geometric identification with star-preserving connections is the real source equivalence, not a claim that arbitrary complex coefficients satisfy an anti-linear star condition. Theorem 7.4 and complex conjugation give $(512 - 204)/2 = 154$ nonreal pairs. □

7.5 An explicit global reconstruction, with safe cancellation

For completeness the other coordinates can be reconstructed without an unspecified interpolation choice. Form a 59×60 coefficient matrix $V(Z)$, with columns indexed by $1, y, \ldots, y^{59}$ and rows

$$f, yf, \ldots, y^{35}f, \qquad g, yg, \ldots, y^{22}g.$$

Let V_j denote its degree-j column and set

$$A_*(Z) = \det[V_1|V_2|\cdots|V_{59}], \qquad B_*(Z) = \det[V_0|V_2|\cdots|V_{59}].$$

If $c_i = (-1)^i \det(V_{\geq 2}$ with row i removed), expanding along the first column gives the constant and linear coefficients B_*, A_*. Every higher coefficient in that row combination is zero by a repeated-column determinant. Consequently

$$A_*(Z)y + B_*(Z) \in (f, g)\mathbb{Z}[y, Z]. \tag{55}$$

The degree bound is $36 \cdot 16 + 23 \cdot 25 = 1151$. All coefficients of both determinants are supplied; equality is checked at 1152 distinct negative integers using the entire fixed-size matrices. Degree drops of specialized input polynomials do not affect these fixed determinant identities.

The exact integer arrays satisfy

$$A_* = c_0 G D_2, \qquad B_* = -c_0 G N, \qquad c_0 \in \mathbb{Z} \setminus \{0\}, \quad \deg G = 388, \quad \deg N = \deg D_2 = 763.$$

Integer convolution checks both identities. Full component tests give $\gcd(\overline{G}, h_i) = \gcd(\overline{D_2}, h_i) = 1$ for every i. At all 512 integral lifts, G, D_2 therefore have nonzero residue. The lifts exhaust $C \otimes L$, so both are units there and descend to units of C. In C, the source minors vanish; cancelling these proved units in (55) gives

$$u_3 = Z, \qquad u_2 = N(Z)/D_2(Z), \qquad u_1 = -r(u_2, Z)/s(u_2, Z).$$

Here $s = -rv = -r/u_1$ is a unit because r and u_1 are units. Formal cancellation in $\mathbb{Q}(Z)$ alone would not have justified these formulas at the fibre points.

The remaining coordinates are given by $m = m_0$, the displayed path formulas, and the unique triangle linear solve. Explicitly, in the unknown order (a, b, c, p, r_1, t_1) the matrix and right-hand side are

$$M = \begin{pmatrix} Y_1 & 0 & 0 & Y_2 & Y_3 & 0 \\ 0 & Y_2 & 0 & -Y_1 & 0 & Y_3 \\ 0 & 0 & Y_3 & 0 & -Y_1 & -Y_2 \\ X_1 & 0 & 0 & -X_2 & -X_3 & 0 \\ 0 & X_2 & 0 & X_1 & 0 & -X_3 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}, \quad w = \begin{pmatrix} -mY_1 \\ -mY_2 - 2Y_1 \\ -mY_3 - 2Y_1 - 2Y_2 \\ -mX_1 - 2X_2 - 2X_3 \\ -mX_2 - 2X_3 \\ -m \end{pmatrix}, \qquad \det M = Y_3 E_1.$$

Cramer's rule gives a literal rational circuit. Its complete 53-node, 30-division serialization is `rational_circuit.json`; its inputs N, D_2 are `u2_reconstruction.json`. Every division denominator is checked to have an explicit inverse in every T_i. Inductively, earlier unit divisions preserve integrality at every lift, and the next nonzero residue proves its denominator is a unit there. Since the lifts exhaust the algebra, each denominator is a unit in C. The exact cofactor relation, the source linear remainder and the unique source reconstruction prove characteristic-zero equality with the original coordinates. Modular residual tests corroborate this implementation; they are not used as a substitute for those exact identifications.

At each real root of H the circuit therefore produces one and only one original real coefficient point, with all 23 factors nonzero. Conversely every original point is obtained in this way. The construction counts marked coefficient points in this localized sector; it does not divide by relabellings or classify a larger moduli space.

A worked real connection. At q^0 with all four scales one, let ρ be the unique root of the primitive polynomial H in

$$-\frac{400542}{10^6} < \rho < -\frac{400541}{10^6}.$$

The exact H-values at the endpoints have negative and positive signs, respectively, and this interval lies inside the certified one-root Descartes leaf $(-1/2, -1/4)$. Set $u_3 = \rho$ and use the existing rational reconstruction circuit. The spatial blocks M_0, M_1 from the original connection ansatz and the scalar have the following certified rounded values:

$$(u_1, u_2, u_3) \simeq (-0.4187284043, -0.4102925777, -0.4005416805),$$
$$m \simeq -1.9869902124,$$

$$M_0 \simeq \begin{pmatrix} 0.757841 & 1.254421 & 0 \\ 0.745579 & -0.088125 & 1.346200 \\ 0 & 0.653800 & 1.317274 \end{pmatrix}, \qquad M_1 \simeq \begin{pmatrix} 0.974629 & 0.528544 & -0.103648 \\ 1.471456 & -0.376875 & 0.689222 \\ 2.103648 & 1.310778 & 1.389236 \end{pmatrix}.$$

For example, the temporal mixed coefficients are

$$(v_{01}, v_{02}, v_{03})(0) \simeq (7.956630, 2.811717, 1.862320),$$
$$(v_{01}, v_{02}, v_{03})(1) \simeq (1.217001, 2.006328, 4.032929).$$

An exact nested dyadic bracket and outward-rounded interval evaluation of the literal circuit certify all displayed rounding. Every one of the original twenty-three units has absolute value greater than 0.1036 on that bracket. The exact algebraic connection is specified by ρ and the rational circuit. These rounded displays illustrate its active spatial and temporal couplings. The supplementary folder `examples/worked_connection/` contains the exact signs, outward enclosures and a separate replay script.

8 Closure of the main theorem

Proof of Theorem 2.1. The scheme-theoretic reconstruction of Section 3 identifies the original declared connection sector with the four-variable family, preserving all inverted factors and all multiplicities. Theorem 5.2 proves the first assertion by a proper closure, complete boundary exclusion at a control point and the actual relative Jacobian criterion.

At q^0, Theorem 7.2 proves that the entire localized fibre is finite. Corollary 6.2 therefore bounds its length by 512. The modular smooth-point construction and its characteristic-zero lifts in Theorem 7.4 supply 512 different points of that same fibre. A point of an Artinian scheme over an algebraically closed field contributes length at least one. Thus the length cap is attained, every point contributes length one, and no extra support or nilpotent contribution remains. This proves the second assertion.

The exact coordinate-algebra identification and Descartes certificate in Theorem 7.7 prove that precisely 204 of these points are real. The rational inverse reconstruction is defined at each point and is over $\mathbb{Q}$, so real primitive coordinates give real coefficient points and conversely. The original real QLC isomorphism then supplies the claimed star-preserving connections. This proves the third assertion. □

9 The computational certificates and reproducibility

9.1 What the computer-assisted part proves

The proof uses explicit finite certificates, grouped by mathematical role. The directory `certificates/geometry/` supplies the original global equations and all cleared localization factors. Its checking programs derive the tensor-component tables, the ring identities and the path and triangle completion identities, and compare the full coefficient arrays with the intrinsic formulas in Section 3.

The additional direct checker `audit/qlc_original_replay.py` imports no other checking module. Starting with symbolic temporal coordinates, all four metric scales and the supported spatial matrices, it reconstructs all 56 coefficient values and expands the original tensor connection and conjugate tensor flip. It checks the explicit ideal representations (22) for all 128 metric and 128 star components, torsion, degree preservation, and both compositions needed for $\sigma^{-1} = D\sigma D$. It also checks the right Leibniz rule on every vertex idempotent at every vertex: $16 \cdot 16 \cdot 4 \cdot 4 \cdot 4 = 16384$ scalar equalities. Two exact negative controls change a reconstructed square coefficient or break an active edge sum; their residual discrepancies are nonzero rational functions. No random or floating-point point test is used. This is a direct check of inverse reconstruction, conditional on the block involution ideal. The separate block certificate proves that the four reduced equations imply that ideal, and the universal-ring argument proves the forward elimination and the identity of the two maps. Neither direction is inferred from the other by a point count.

The directory `certificates/generic/` supplies finite-field coefficients and unit identities for every finite boundary chart and pole chart, the restricted elimination data, the 512-degree control algebra, and the relative Jacobian tests. Their role is to check the finite algebraic calculations in the generic-degree proof; the properness and specialization arguments are proved in the text.

The directory `certificates/centre/` contains all numerator polynomials used in the quadratic-module reduction, all six minors, the eleven smooth residue components, the full integer eliminant and its factorization, the primitive coordinate polynomial, the complete Descartes partition, and the rational reconstruction circuit. These are the actual coefficient arrays, not only their checksums. Every division that occurs in a coordinate reconstruction is justified by a unit condition or an explicit inverse certificate. The proof does not equate a determinantal plane quotient with the original connection scheme, and does not discard a rank-zero locus without an argument.

For a long polynomial identity over $\mathbb{Z}$, exact evaluation and interpolation are a proof when the declared degree bound is justified and the number of distinct exact test points exceeds that bound. The eliminant and cofactor checks use grids disjoint from the production grids. A handful of additional determinant samples is only corroboration. Finite-field unit statements are verified by explicit products equal to one, or Bézout identities, in the specified quotient algebra. No floating-point residual is used to prove an exact equality.

9.2 How to read or replay the proof

The package root contains `manuscript.tex`, `manuscript.pdf`, the complete standalone `supplement.tex` and `supplement.pdf`, `audit/`, `certificates/`, `requirements.txt` and the portable worked example in `examples/worked_connection/`. The common counting exposition and geometric consequences are identical in both TeX sources, including the text between formal statements and proofs. The files in `sections/` are retained literal source-binding inputs, not verbatim extracts of the current printed sections; the complete documents are the canonical

exposition. The checker `check_consistency.py` compares these complete regions, checks all references and cited package paths, and binds both sources to this release. No separate historical counting-proof document is required.

After installing `requirements.txt`, run

```
python -B reproduce.py --output ../qlc-results
```

The output directory must be new and outside the package. The entry point first checks source and package consistency, then runs `verify.py` in both normal and explicit Python `-O` modes, and finally replays the worked example with outward-rounded interval arithmetic. The counting runner stages only 175 catalogued proof inputs, including 142 coefficient files, in a fresh temporary workspace. Prior outputs and review reports are never proof inputs. Each child must exit successfully and create a new receipt with its exact full-scope result status. The complete determinant checks cannot be replaced by supplemental samples.

The supplement reports complete normal and optimized executions, their receipt hashes, and the bytewise comparison of all mathematical result pairs. The stated environment is Python 3.12.14, SymPy 1.14.0, python-flint 0.9.0 and native FLINT 3.6.0.

An exact cross-block join binds all nine graph equations, all twenty-three localization factors and both triangle invariants across the source and degree calculations. Additional literal-formula checks bind the rational circuit and the compact-top scalar normalization. With $T_0 = -P_{\mathrm{top}}|_{q_*}$, equation (37) has scalar -4; using the raw top polynomial instead has scalar $+4$. Both scalar identities are checked in characteristic zero.

Python integers supply direct polynomial and quotient arithmetic. SymPy and FLINT are explicit arithmetic dependencies for the specified normalizations and large exact determinants. Hashes identify coefficient bytes; executed identities and the written arguments establish the mathematics. Scheme isomorphisms, properness, the finite-fibre length lemma, Hensel lifting and faithful descent are proved in the text and are not inferred from a software status label.

9.3 Use of generative AI in the research workflow

ChatGPT and OpenAI Codex assisted exploration of the algebraic reductions, proof drafting, literature triage, development of checking programs and internal adversarial reviews. The deterministic certificate checks use the explicit coefficient arrays and algorithms described above. Their input catalogue, exact outputs and executable replay are supplied in full, so independent reproduction requires the stated arithmetic environment and does not require a generative model or an earlier conversation. The scheme identification, properness, length and lifting arguments are written mathematical arguments requiring critical review. Internal agent reviews contributed to checking the preparation; they are separate from the exact arithmetic certificates and from external journal refereeing.

9.4 Scope of the theorem and of the review

The family is restricted by its chosen calculus, spatial homogeneity, ordered path and triangle, trace constraints and the exact regular open. These conditions are part of the definition. They have not been proved to exhaust all connections on the cube. Nor is the number divided by a relabelling group or interpreted as a count of physically inequivalent geometries.

The result is the consolidated degree and fixed-metric count. No explicit parameter-radius optimization, quantum functional integral, regulator removal or physical continuum claim is needed for it. The number 204 is obtained only after the complexified coordinate algebra and its real structure have both been identified.

10 Conclusion

The declared path–triangle QLC family admits an exact reversible algebraic model. The corrected boundary and degree argument, finite-fibre length bound and smooth-lift exhaustion establish a complete reduced 512-point fibre, not merely a modular lower bound. Its primitive coordinate algebra and exact real-root certificate give 204 actual star-preserving connections at the stated rational metric.

Corollary 10.1 (Distinct tensor Leibniz maps at one metric)**.** *At q^0 with scales one, the* 204 *counted real connections have* 204 *distinct tensor Leibniz maps σ.*

Proof. For this calculus the translation-degree argument in Section 3 forces the inner correction map γ to vanish. Hence $\nabla\omega = \theta \otimes \omega - \sigma(\omega \otimes \theta)$. Two equal maps σ would give equal connections and equal coefficient points, contrary to the distinctness already proved. □

This observation locates the finite multiplicity in the varying bimodule data; it does not conflict with a comparison fixing the same σ. Classification of excluded support or denominator strata, special metrics and symmetry quotients are further questions. The enumeration supplies no curvature formula, action principle or physical selection of this support.

Corollary 10.2 (Obstruction to ordinary coordinate products)**.** *None of the* 204 *real connections counted at q^0 is an ordinary tensor product of connections for a nontrivial partition of the four coordinate directions. This remains true after permuting those direction labels.*

Proof. For an ordinary product over a partition, each frame element e^i belongs to one factor and ∇e^i has both output directions in that factor. Therefore $C^i{}_{jk} = 0$ whenever i, j, k do not all lie in a single block. Take the block containing direction 0. A nontrivial partition has some spatial direction a in another block, so it would require $C^0{}_{0a} = v_{0a} = 0$. On our declared open, however, (9) gives

$$v_{0a}(0) = \frac{q_a - d_a^2}{q_a - d_a}, \qquad v_{0a}(1) = \frac{q_a - d_a^2}{d_a(q_a - d_a)},$$

and every displayed numerator and denominator is a unit. These values cannot be zero at a real point. The three active pairs $\{0, a\}$ form a connected graph on all four directions; connectedness and the same obstruction survive any permutation of the labels. □

The statement concerns ordinary products of the specified coordinate calculi. It excludes neither a more general extension construction nor an alternative change of geometric data.

A An explicit comparison with Aschieri–Weber compatibility

This appendix supplies the tensor calculation behind the same-metric comparison with [22, Definition 5.3(ii), Theorem 5.7]. Its conventions and hypotheses are retained: it is not a comparison obtained merely by fixing our unknown tensor Leibniz map σ.

Put $A_G = \mathbb{C}(G)$, $G = (\mathbb{Z}/2)^4$, and $k_i = h_i^{-1}$. Edge symmetry gives $R_i k_i = k_i$. For the right-linear dual frame ξ_i, with $\langle \xi_i, e^j \rangle = \delta_{ij}$, the corresponding inverse metric is $\mathfrak{g} = \sum_i k_i \xi_i \otimes_{A_G} \xi_i$. This calculus is bicovariant; the canonical Woronowicz map σ_W flips its invariant frame tensors, and the exterior relations are Grassmann relations. This canonical map is distinct from the variable σ of our bimodule connections.

The rational tensor product. Let ϕ_0 be a right-linear coefficient map constant on invariant frames and let $\phi = f\phi_0$, for any $f \in A_G$. Write S_H for the Hopf antipode and use Sweedler indices for the iterated coproduct of f. The internal-Hom coaction and rational tensor formula [22, equations (2.2), (2.6), (2.11)], on invariant inputs, give

$$(\mathrm{id} \otimes_{\sigma_W} \phi)(e^i \otimes_{A_G} e^j) = \sum f_{(1)} e^i \otimes_{A_G} S_H(f_{(2)}) f_{(3)} \phi_0(e^j).$$

Moving the second coefficient through e^i yields the coefficient

$$\sum f_{(1)} R_i\Big(S_H(f_{(2)}) f_{(3)}\Big) = f, \qquad \sum f_{(1)} \otimes S_H(f_{(2)}) f_{(3)} = f \otimes 1.$$

The second equality is the antipode identity and proves the first. Thus the internal-Hom contribution and tensor balancing cancel the apparent coefficient shift for every vertex function, without a covariance or spatial-homogeneity restriction on f. For connections, the rational tensor product in [22, equation (2.23), Section 4.4] reduces on invariant frames to the same calculation: split off the canonical connection, which vanishes on those frames and their tensors, and apply the calculation to the remaining right-linear map.

Contracting with $\mathfrak{g}$ on $(e^i \otimes e^j + e^j \otimes e^i)/2$ now gives the actual specialization of [22, equations (5.9)–(5.10)]:

$$M_{ijb} = \Phi_{\mathfrak{g}}(L)_{ijb} = k_j L^i{}_{jb} + k_i L^j{}_{ib}, \qquad L^i{}_{jb} = L^i{}_{bj}, \quad M_{ijb} = M_{jib}.$$

Its inverse is

$$L^i{}_{jb} = \frac{k_b M_{ijb} + k_j M_{ibj} - k_i M_{jbi}}{2 k_j k_b}.$$

Substitution verifies both compositions; the permuted cross terms cancel and the remaining numerator is $2k_j k_b L^i{}_{jb}$. All denominators are metric units. Consequently $\Phi_{\mathfrak{g}}$ is invertible here, including on arbitrary coefficient functions, and the hypotheses of their Theorem 5.7 apply.

The compatibility predicate and the left/right passage. Write $\nabla_R e^i = \sum_{a,b} K^i_{ab} e^a \otimes_{A_G} e^b$. The same tensor calculation specializes their metric condition to

$$k_j K^i_{jb} + k_i K^j_{ib} = \delta_{ij}(R_b k_i - k_i). \tag{56}$$

Under $\nabla_R = D \circ \nabla \circ *$, the star and tensor flip of Section 3 give

$$K^i_{ab} = -R_a R_b(\overline{C^i{}_{ba}}), \qquad K^i_{ii} = -\overline{U_i}.$$

Setting $i = j = b$ in (56), and using $R_i k_i = k_i$, forces $U_i = 0$ for all four directions. This conflicts with the imposed trace $\sum_i U_i(t) = 4$ in (5).

The difference also persists without that trace restriction. Put $Q_i(t) = h_i(1-t)/h_i(t)$ for $i > 0$. The inverse formula above gives

$$K^i_{i0}(t) = \frac{k_i(1-t) - k_i(t)}{2k_i(t)} = \frac{Q_i(t)^{-1} - 1}{2}, \qquad K^0_{0i} = 0.$$

Pulling back shifts the first coefficient to the opposite time slice; hence $v_{i0} = (1 - Q_i)/2$ and $v_{0i} = 0$. Reconstructing the corresponding σ by (2) in the original tensor metric equation (3) yields

$$\frac{E_{0ii}}{h_i} = (Q_i - 1) + 2v_{i0} - v_{i0}^2 = -\frac{(Q_i - 1)^2}{4} \neq 0. \tag{57}$$

Here $Q_i - 1$ is a unit on our metric-ratio base. Thus the unique connection for the Aschieri–Weber predicate does not satisfy our original full tensor metric predicate under this same-metric left/right identification. This establishes the stated comparison; it excludes no arbitrary alternative realization of the geometric data.

The standard-library checker `literature/comparison_checks/aw_comparison_check.py` recomputes both inverse compositions at q^0, the rational-tensor identity on every vertex basis function and direction, and the six normalized residuals on the two slices. Its optional `-source-pdf` argument identifies the inspected arXiv v2 bytes; it is not needed for the arithmetic. This bounded comparison check is separate from the complete 512/204 counting replay. The general identities and their geometric interpretation are the written derivation above, not an inference from a software status.

Data availability. The exact coefficient data, verification programs, certificates and execution records are publicly available in the versioned GitHub research package [23], at commit `1cefff4`. The complete code/data archive accompanies article revision 0.13 and supplement revision 2.3. It contains the same 175 protected counting inputs as the delivered supplement. The current package also includes the complete editable sources, pinned requirements and exact replay records. Run `reproduce.py`; `inputs.json` and `synchronization.json` identify and bind the proof inputs and sources.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript and its computational supplement, the author used ChatGPT and Codex (OpenAI) for drafting and revising the text, mathematical support, and internal AI-assisted review of the arguments and computations. Further details are given in the reproducibility section. The author previously reviewed the manuscript and supplement, authorized the subsequent documentation revisions, and takes full responsibility for the content.

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